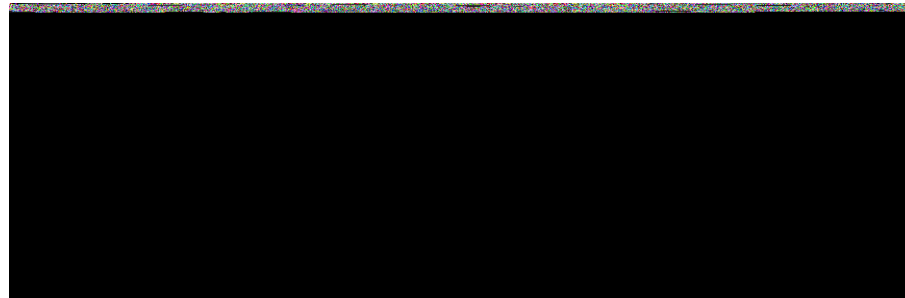


Efficiency of Microstructural Reinforcement for Concrete in Tension

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C3 Symposium 2025, Chicago, IL, USA,
October 2nd-5th, 2025



The Hypothesis: Are we doing enough?

#1: Changing Societal Attitude towards Concrete, What will the next generation ask?

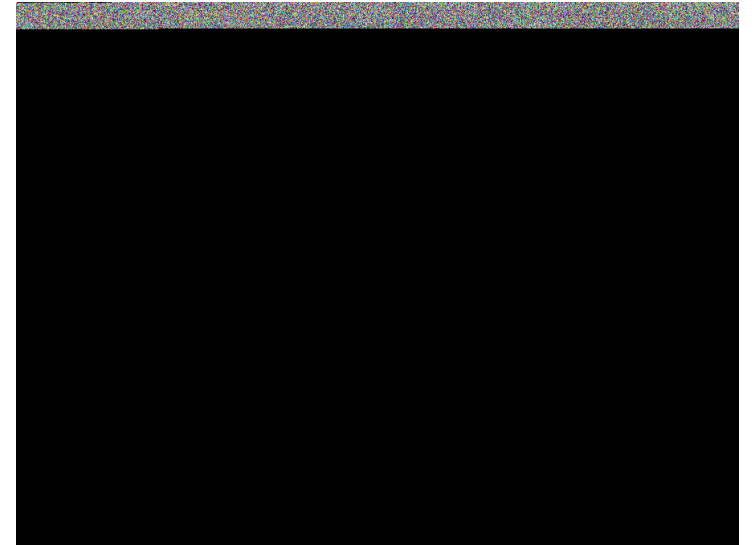
#2: Reinforcement Strategy is the Cornerstone of Concrete consumption, what is being done?

- China's Cement use in 3 years > US use in 100 years [6.6 GT (2011-2013) vs. 4.5 GT (1901-2000)]
- About 60-75% concrete under flexure doesn't carry any load due to elimination from structural calcs
- Very little has been done about efficient structural design of concrete



Guardian, Best of 2019,
Concrete: the most destructive material on Earth

<https://www.theguardian.com/cities/2019/feb/25/concrete-the-most-destructive-material-on-earth>

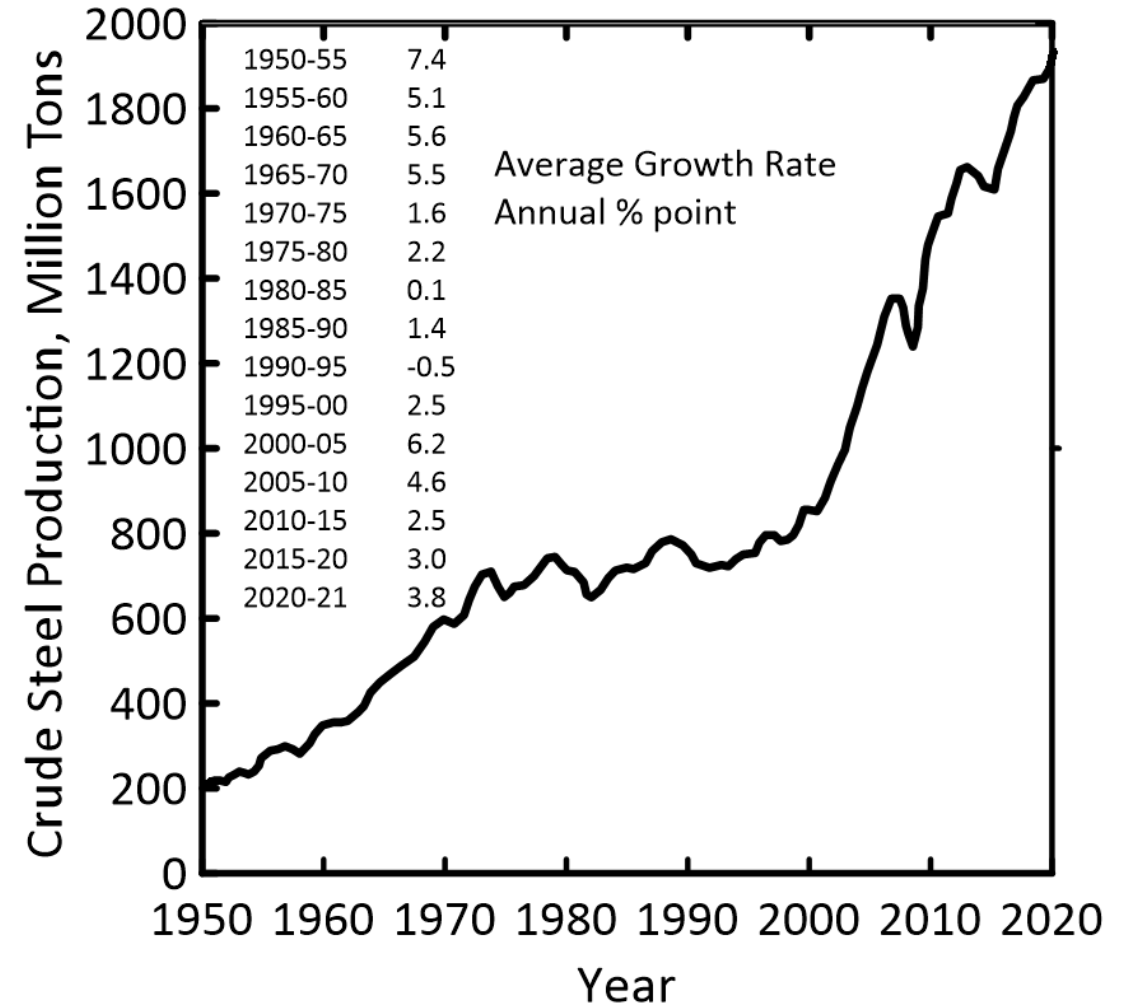


Scientific American, March 1, 2024,
The False Promise of Carbon Capture as a Climate Solution

<https://www.scientificamerican.com/article/the-false-promise-of-carbon-capture-as-a-climate-solution/>

World Production of Steel in the past 70 years

- Crude steel production has increased from 800 to 2000 million tons per year since 1990
- One-half of the current production of 1 billion tons a year is by China, 10 times more than India, the second producer.
- assuming annual concrete consumption of 10Bt Total volume of rebar will be 522 million tons
- 7.5 lbs of steel rebar per ft³ of concrete
- Although recyclable, the raw feed supplies of steel ore will be exhausted due to increasing demand.

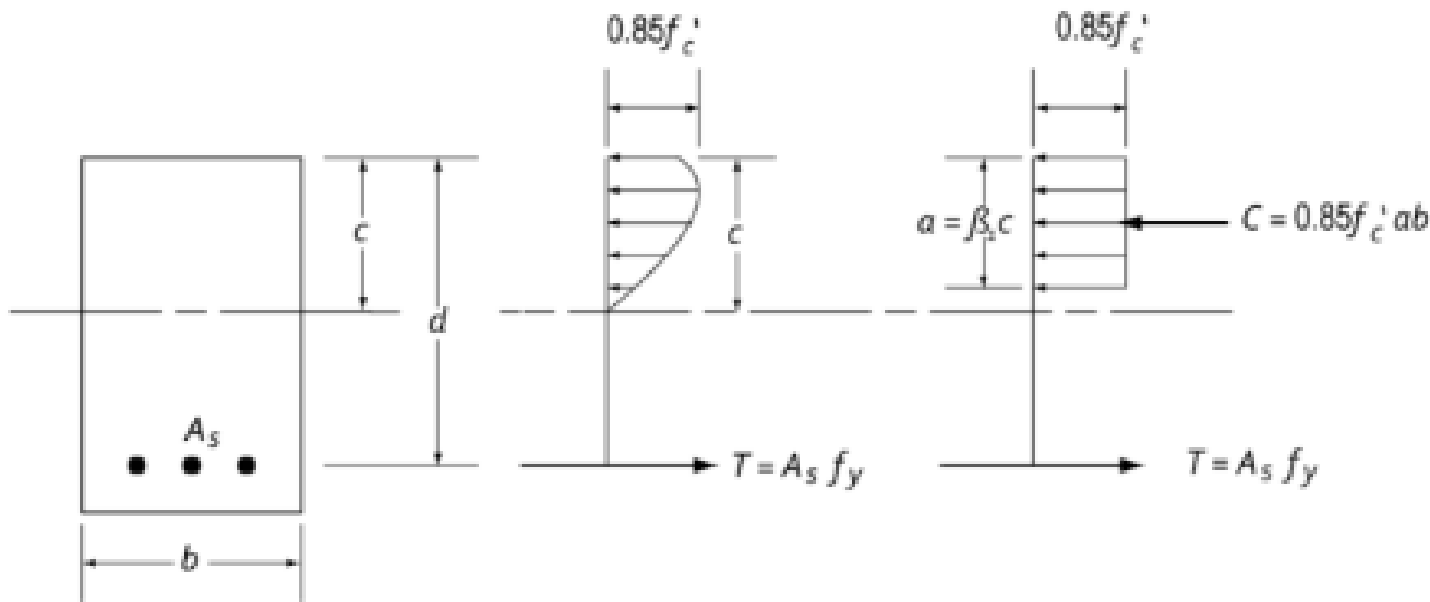


Strategies to Improve The Efficiency of Structural Concrete

- Efficiency Defined through Rational quantification of carbon footprint
 - Stress to strength ratio for both tension and compression,
 - Normalized Specific Strength for competing systems
 - Specific Strength per volume
 - Compressive strength is a poor measure Kg CO₂/MPa
- Proposed Solutions:
 - Fiber, Textile, Hybrid and UHPC Reinforced Concrete
 - Applications addressing min. reinforcement
 - Repair and New construction applications/products
 - Structural design fibers, Precast sections, minimum reinforcement, tunnel lining, Slabs, Environmental Applications
 - Design Equations for Resilient and Ductile Systems
 - SLS, Combination rebars and fibers, TRC sections, UHPC, Structural FRP sections

Concepts for the structural design are more than 120 years old

- Structural Design of reinforced concrete has been practically unchanged in 120+ years
- Guidelines developed in 1907, modified in 1930s, and codified in 1940s.
- Approximately 60-70% of concrete's volume that is subjected to tensile stress is still ignored and thus a waste of resource, providing no contribution other than providing a cover for the steel rebar.



Rectangular Stress Block
Charles S. Whitney, 1930s

$$M_u = \phi A_s f_y d \left(1 - 0.59 \frac{\rho f_y}{f'_c} \right)$$

Formulas.

Position of neutral axis,

$$k = \sqrt{2n \left(p + p' \frac{d'}{d} \right) + [n(p + p')]^2} - n(p + p') \quad (26)$$

Position of resultant of compressive stress, $C + C'$,

$$x = \frac{k + \frac{d'C'}{dC}}{\frac{C'}{C} + 1}; \text{ in which } \frac{C'}{C} = \frac{2p'n \left(k + \frac{d'}{d} \right)}{k^2} \quad (27)$$



FIG. 67.

Arm of resisting couple,

$$j = \left(1 - \frac{x}{d} \right) \quad (28)$$

Moment of resistance,

$$M_s = f_s p j \cdot b d^2 \quad (29)$$

$$M_s = \frac{1}{2} f_c k (1 - \frac{1}{2} k) b d^2 + f'_s p' b d (d - d') \quad (30)$$

Fibre stresses,

$$f_s = \frac{M \div j d}{A} \quad (31)$$

$$f_c = \frac{k}{n(1-k)} \cdot f_s \quad (32)$$

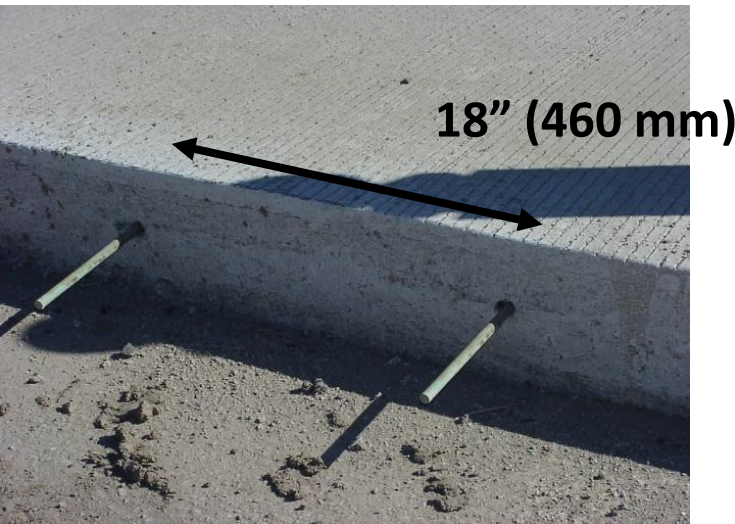
$$f'_s = \frac{n \left(k - \frac{d'}{d} \right)}{k} \cdot f_c = \frac{k d - d'}{d - k d} \cdot f_s \quad (33)$$

Ultimate Loads formula by Tuneau 1907

Why Should we move towards Supplementary Reinforcement?

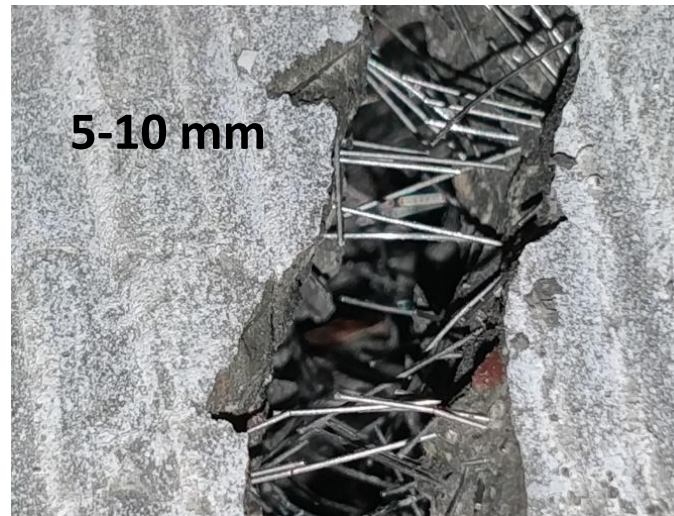
Strengthening at the microstructure level will have a profound impact at the macro-structural level strength and durability

Concrete reinforcement at a microscale is a game changer in terms of Carbon footprint



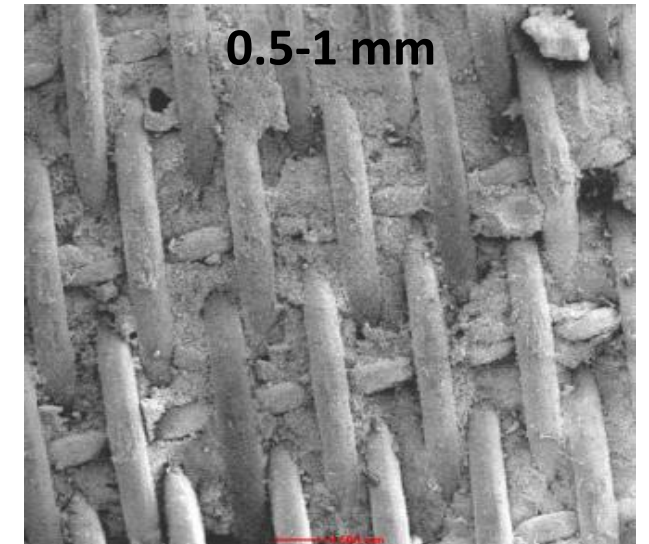
Ordinary reinforced concrete slab, 5-10" thick, standard rebars, 12"-18" apart

Reinforcing Bars: Macro-structure reinforcement



Fiber reinforced Concrete (FRC), at the meso-structure with $\frac{1}{2}$ " long fibers, $\frac{1}{2}$ " (10mm) millimeters apart

Fiber reinforced Concrete: Meso-structure reinforcement



Textile reinforced Concrete (TRC), at the micro-structure with long fibers, woven, only millimeters apart

Textile Reinforced Concrete: Micro-structure reinforcement

Delayed Activation of Reinforcement

Inhibited Localization

What are the potential mechanisms for toughening Concrete?

■ Mechanisms:

- Crack path tortuosity
- Crack deflection
- Crack bridging, fiber pullout, yielding
- Delamination cracks
- Multiple cracking, Tension stiffening

■ Structural Redundancy in the Limit States

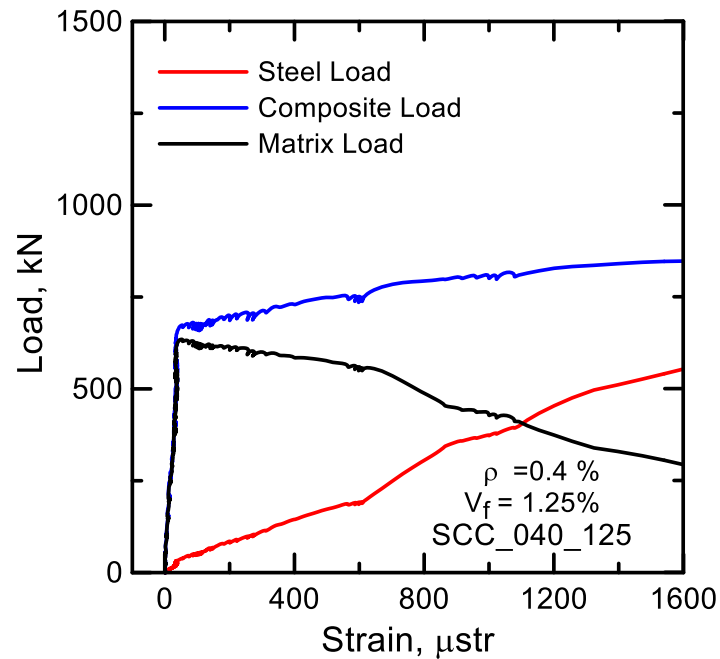
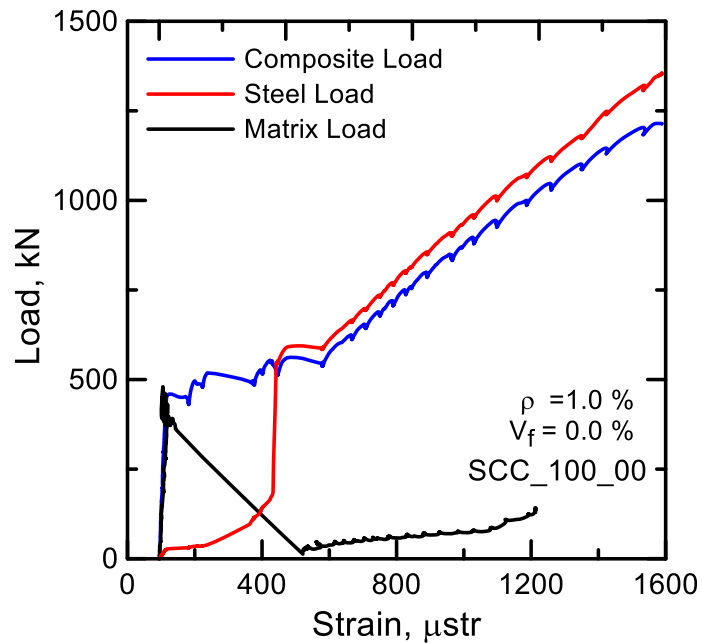
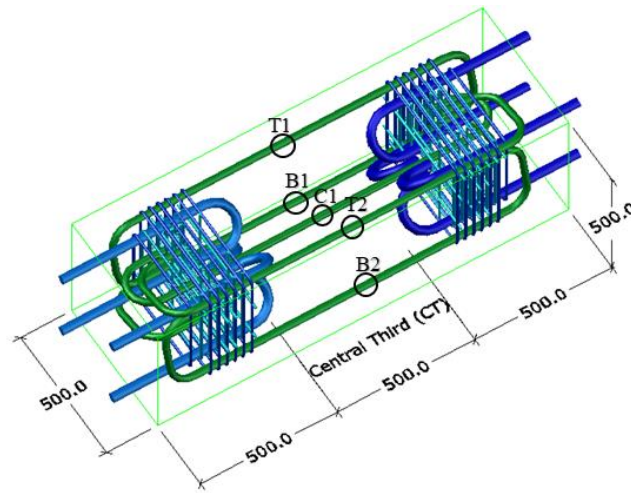
■ Correlation of cracking with creation of new surfaces and energy dissipation

■ Delay localization to increase the strength of concrete in tension

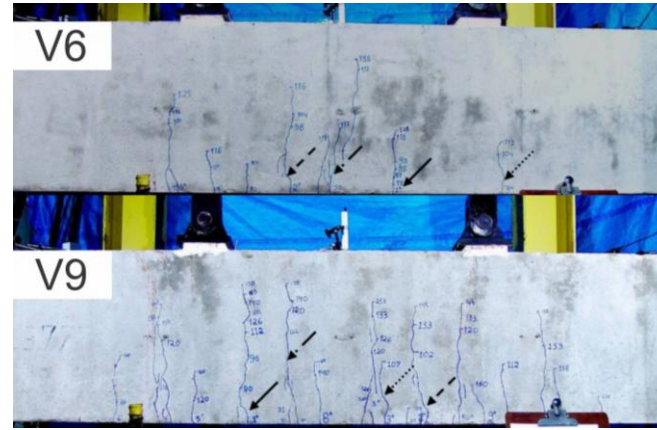
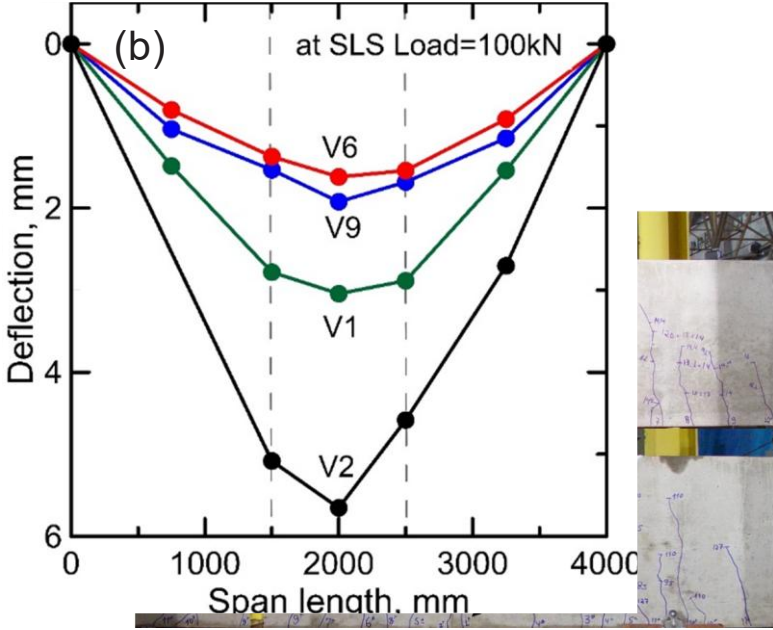
■ Delays in crack propagation, reducing w/c , porosity, and improving interfaces

■ Transverse Yarn anchorage by textiles

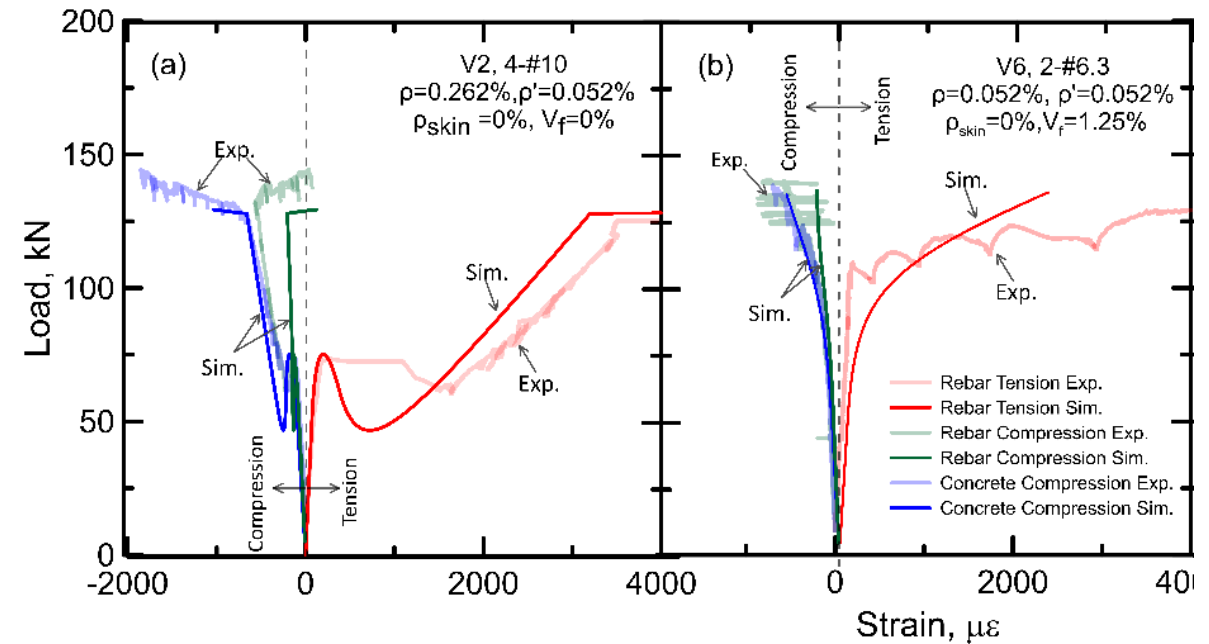
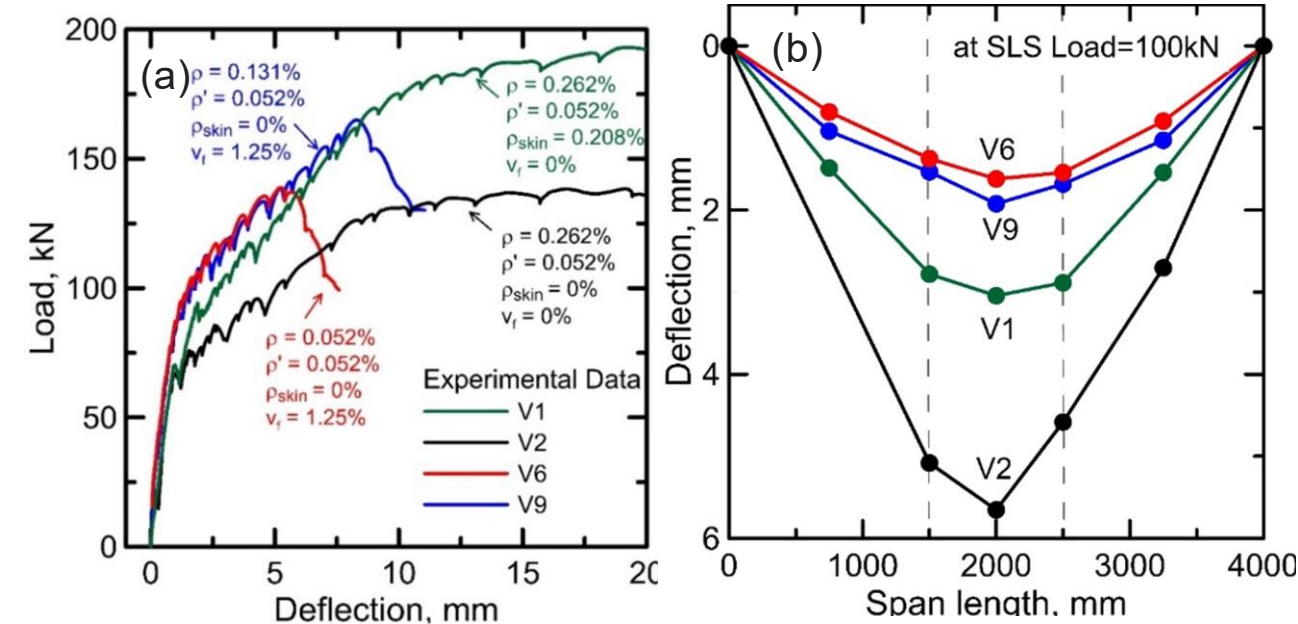
Tension Member



Effect of steel fiber reinforcement on the energy dissipated during loading
(average strain of 10 strain gages is shown) a) No fibers, 1% rebar. (SCC_100_00)
b) 1.35% Steel fibers, 60% reduction of Rebars

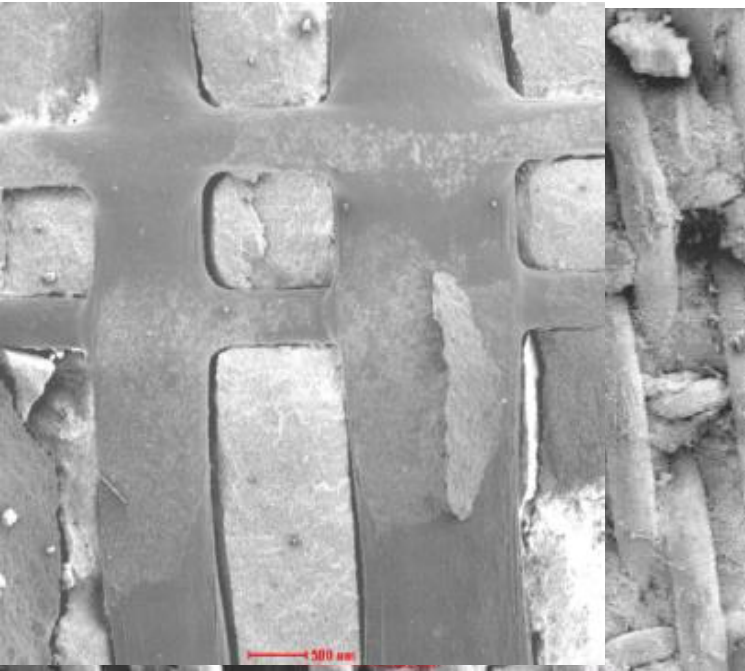


Comparison the crack pattern of plain reinforced and Hybrid reinforced concrete systems at rebar strain ($\epsilon_{st} = 0.01$).

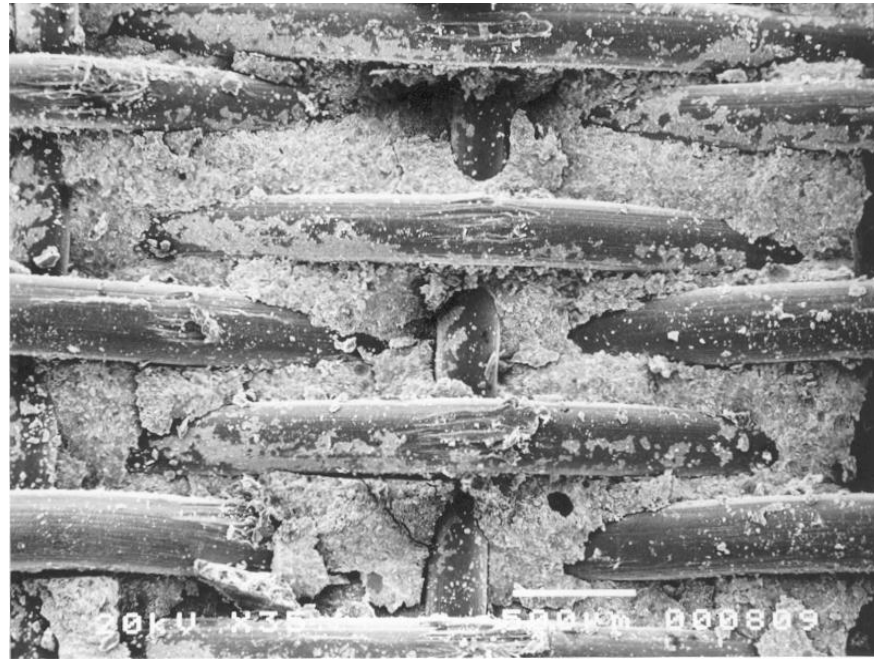


load-strain curves between analytical simulations and experimental data for concrete compression, rebar compression, and rebar tension

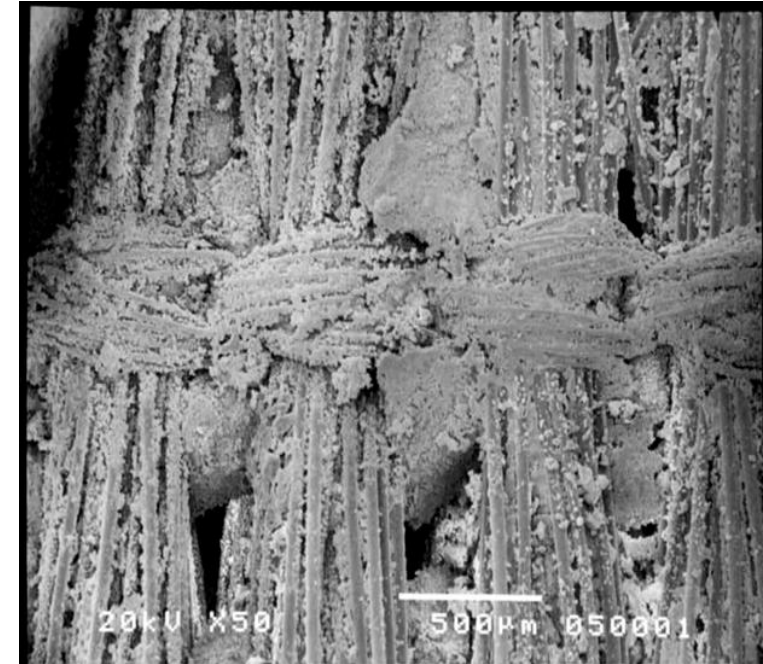
Fabrics in Paste- Toughening as a scale issue



AR Glass
Bonded Fabric

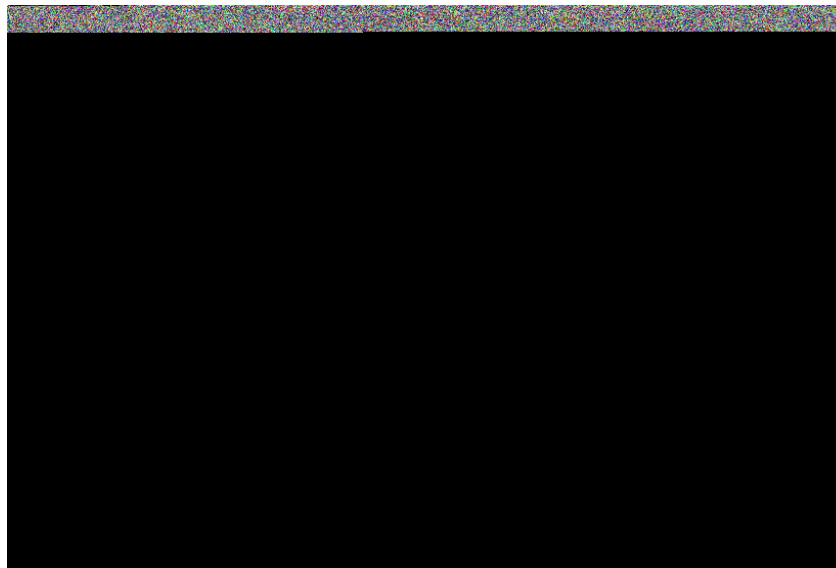
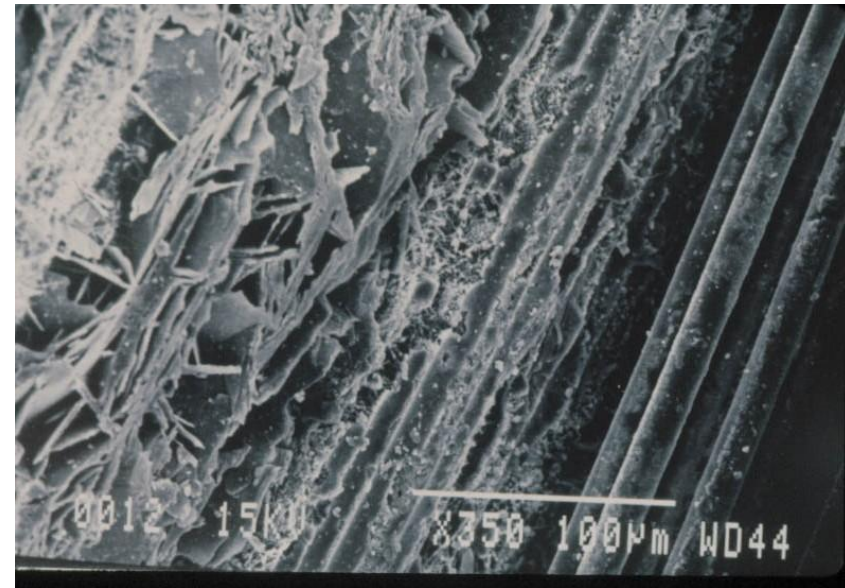
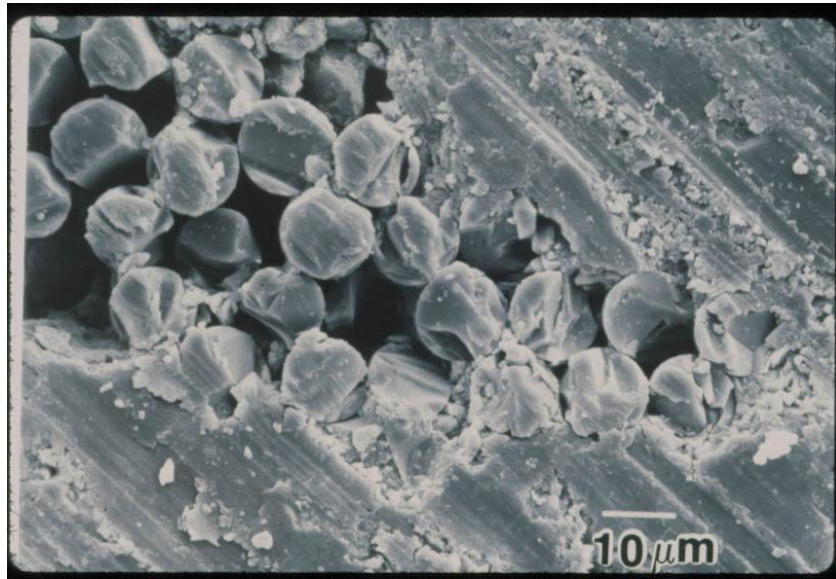


Polyethylene (PE)
Woven Fabric

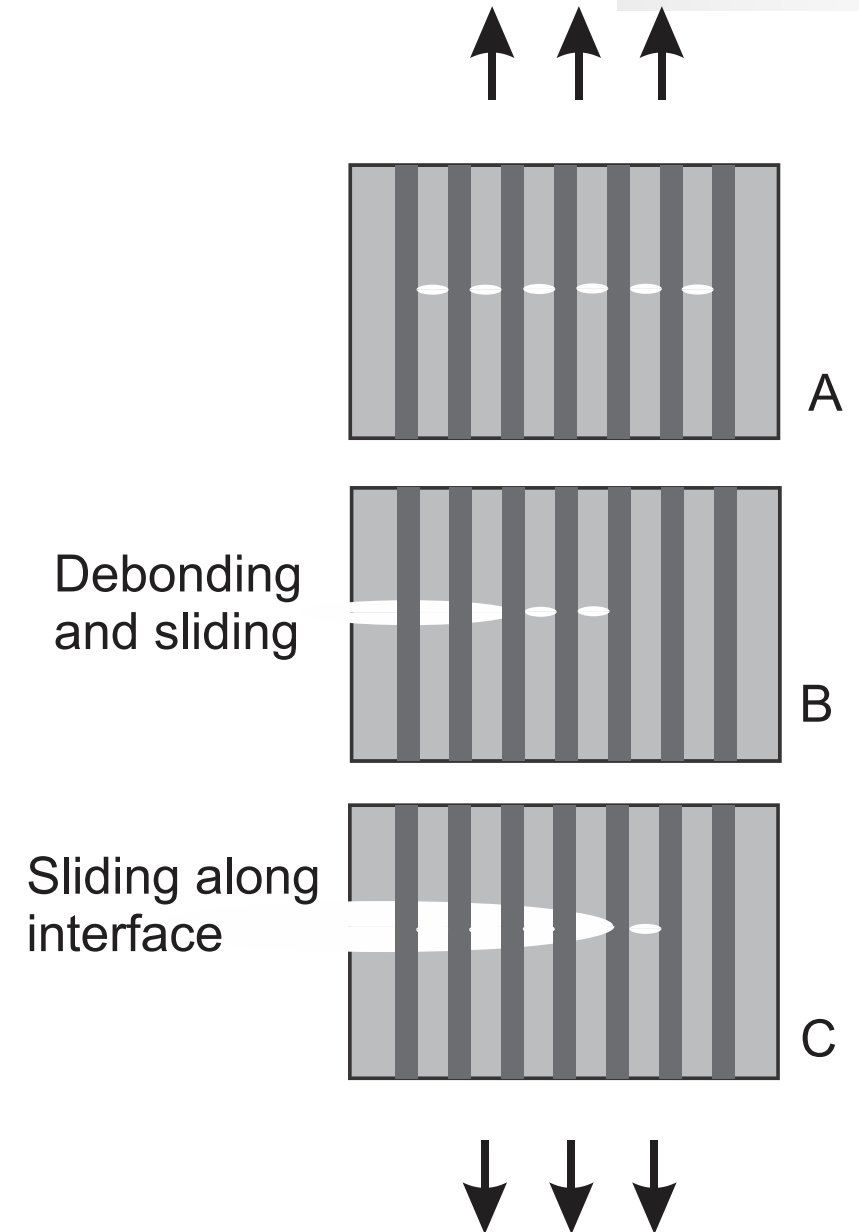
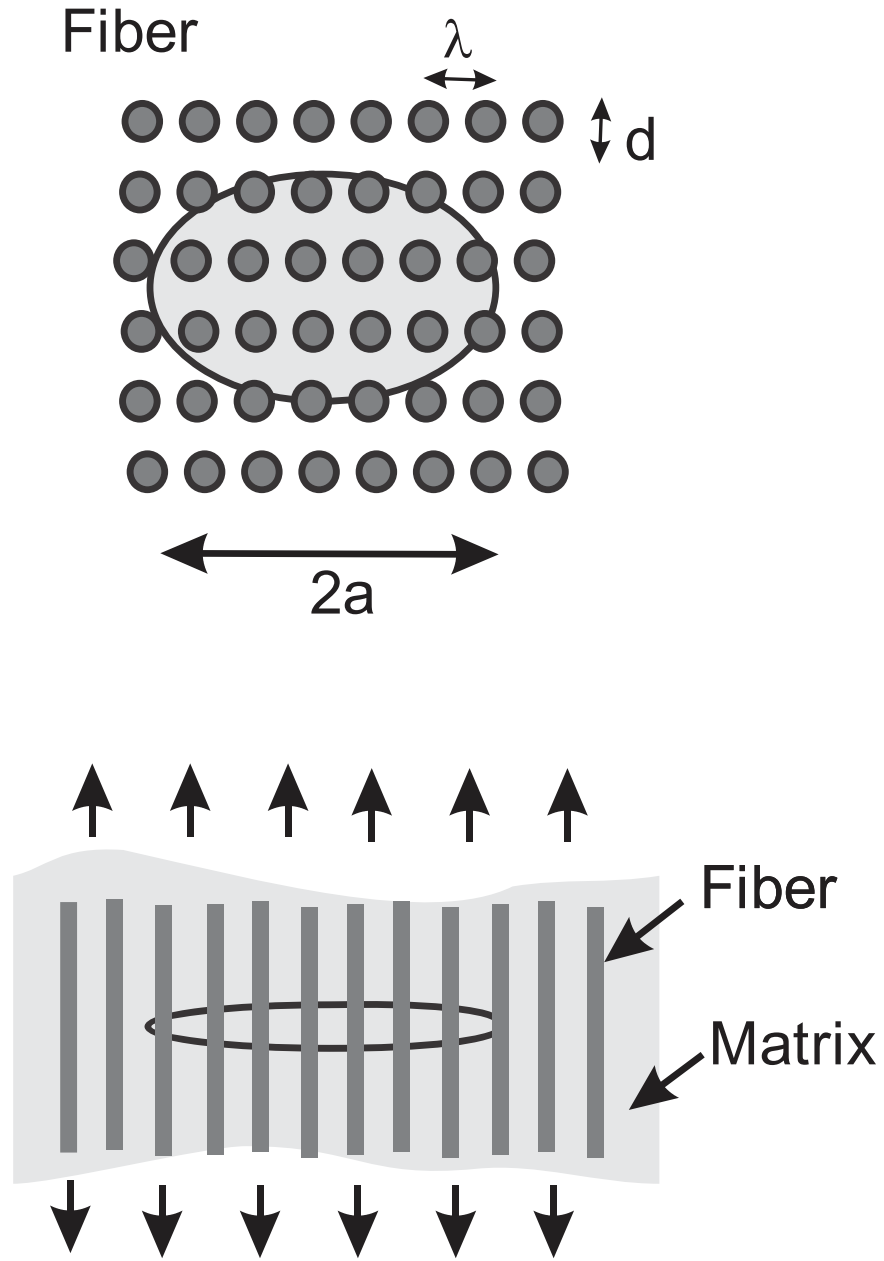


Polypropylene (PP)
Knitted Fabric

Interface Transition Zone



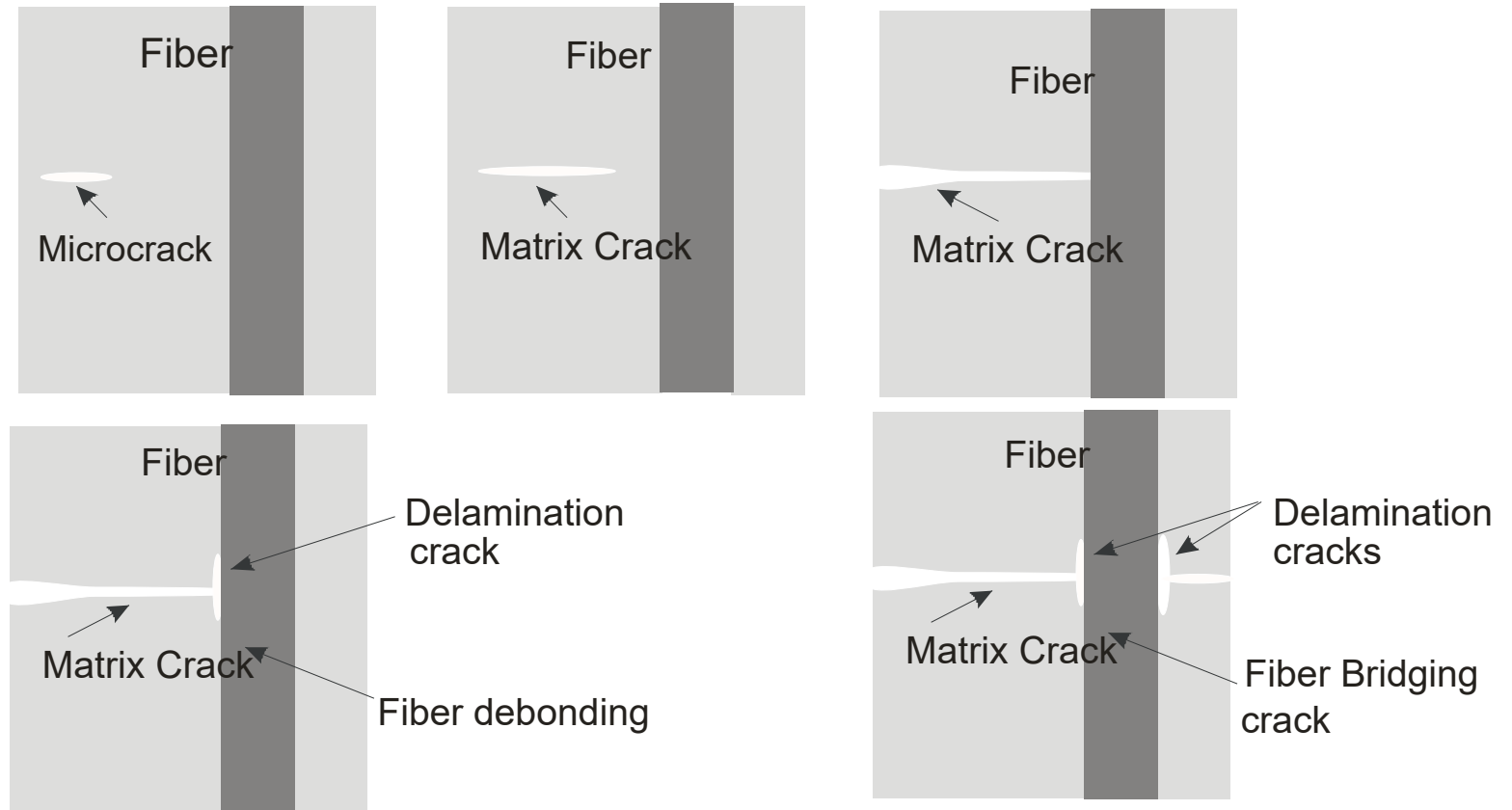
First Cracking Stage- Inhibited Localization



Sequence of cracking from initiation and growth, deflection, & Bridging perspective

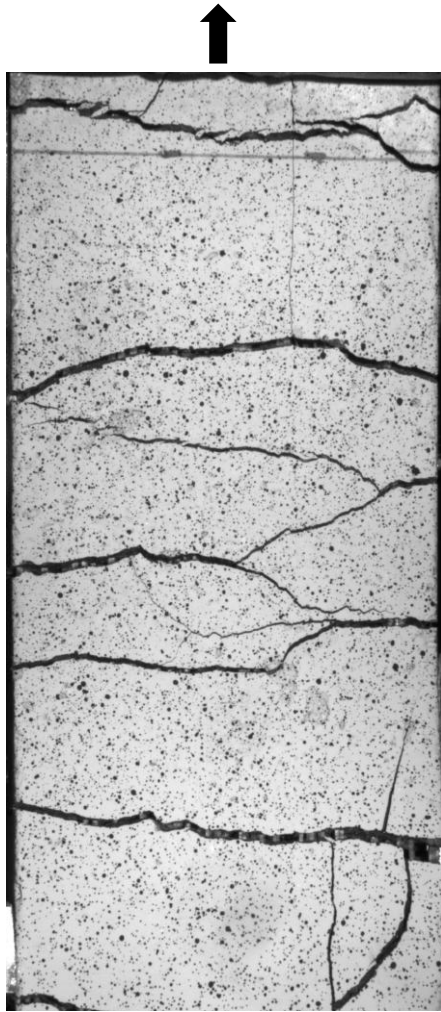
The cracking process starts from slow crack growth of a microcrack from a pore or aggregate interface, crack propagation, hindered by fibers, resulting in crack deflection, delamination, fiber bridging, pullout

1. Stable microcrack to macro crack growth
2. Crack coalescence and growth
3. Crack arrest by fibers
4. Fiber debonding and crack deflection
5. Interface debonding
6. Full fiber debonding growth
7. Fiber bridging

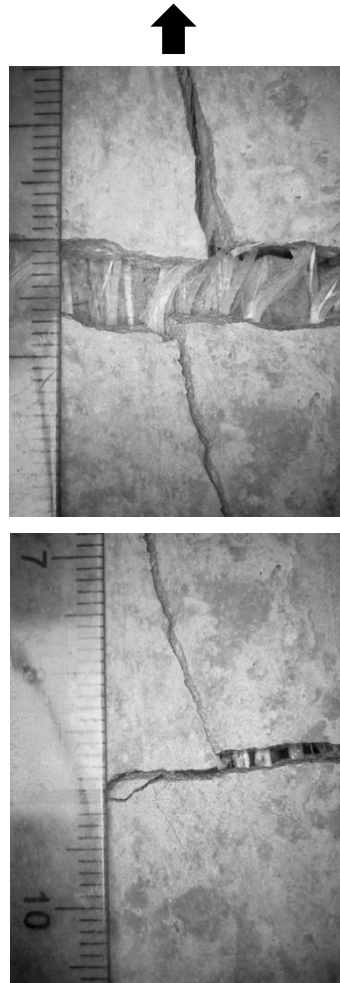


All these mechanisms are absent when concrete is unreinforced at the microstructural level

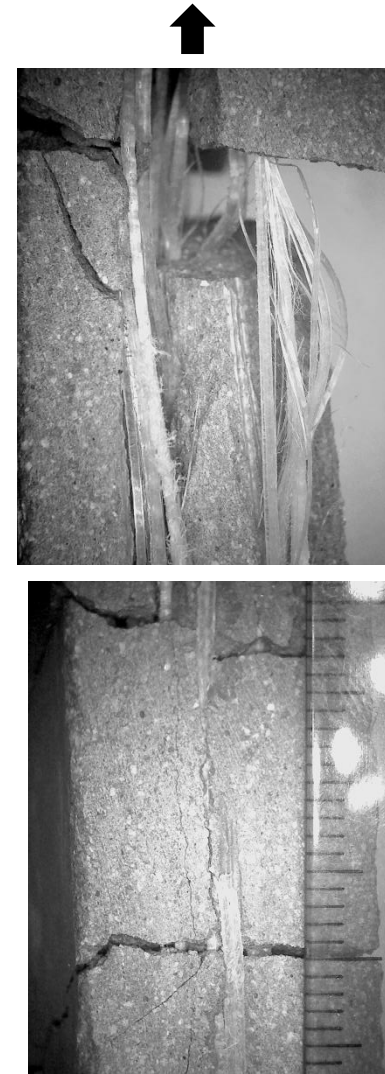
Toughening Mechanisms



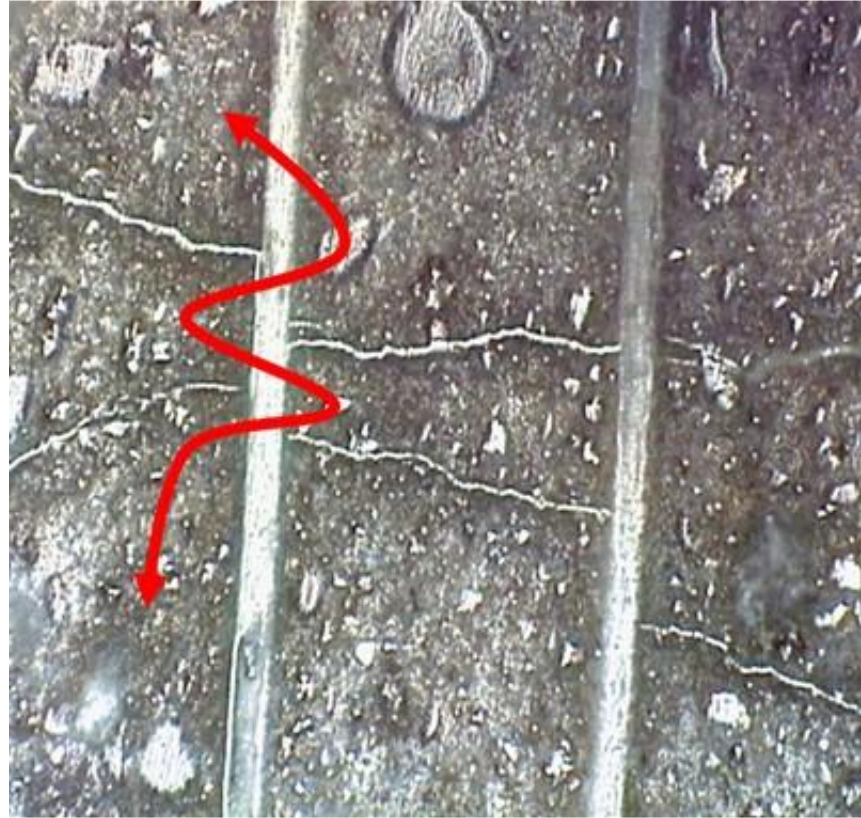
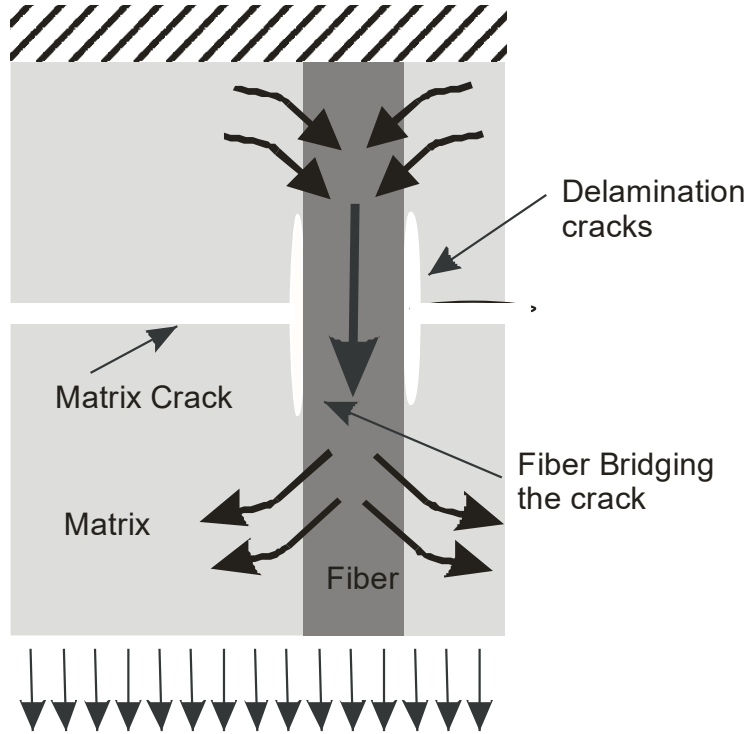
Distributed cracks across loading direction



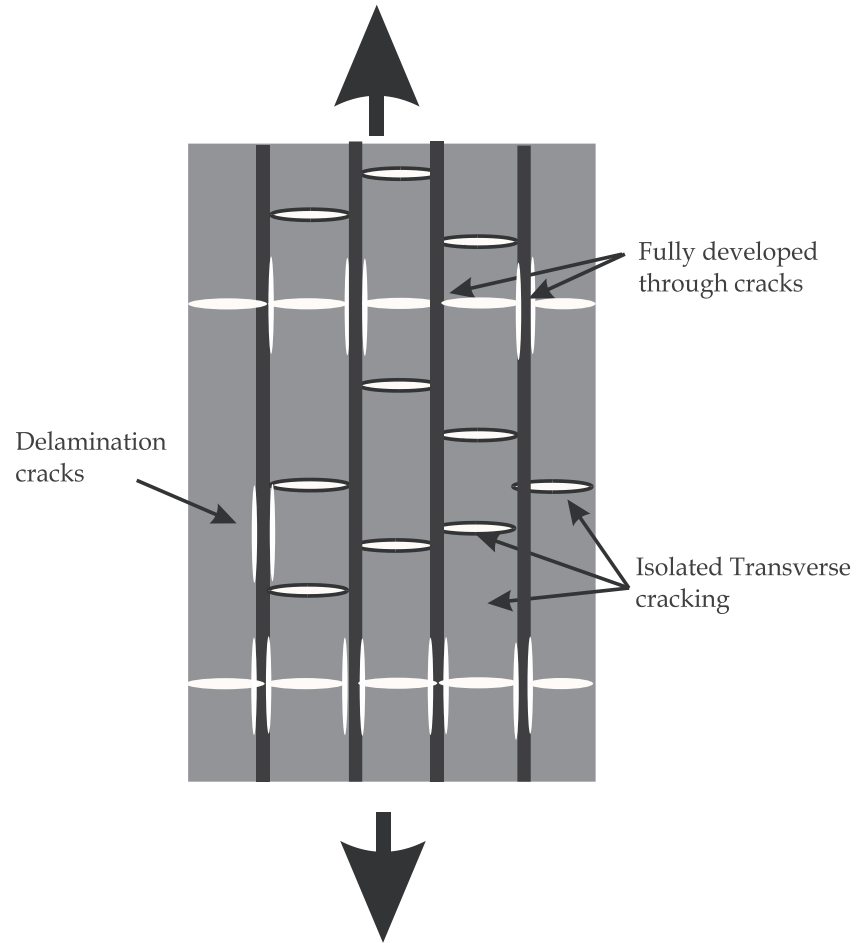
Fiber bridging across loading direction



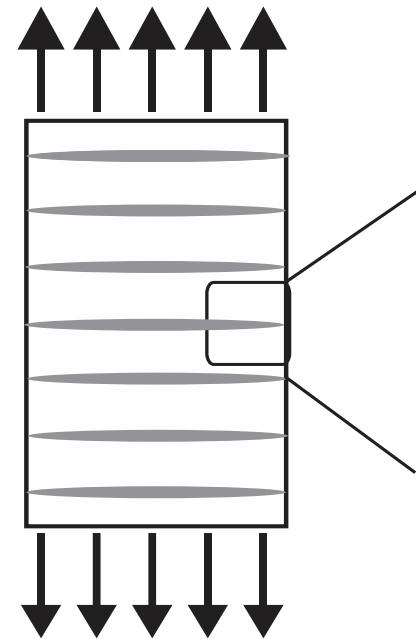
Tension Stiffening: major mode of toughening at the structural level



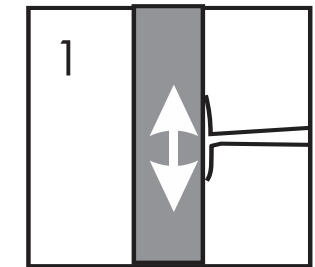
Micro Toughening Mechanism



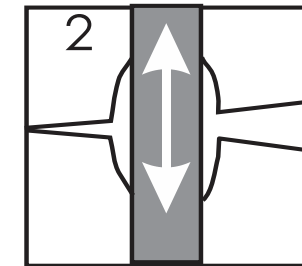
Fibers and fiber-matrix interface prevents complete localized failure in the matrix place through a series of distributed cracks transverse to the direction of the load.



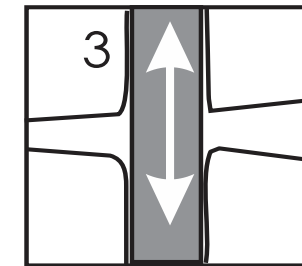
Distributed cracks enable deflection of matrix cracks through fiber-matrix debonding and frictional sliding of the fibers under tension



Crack Deflection

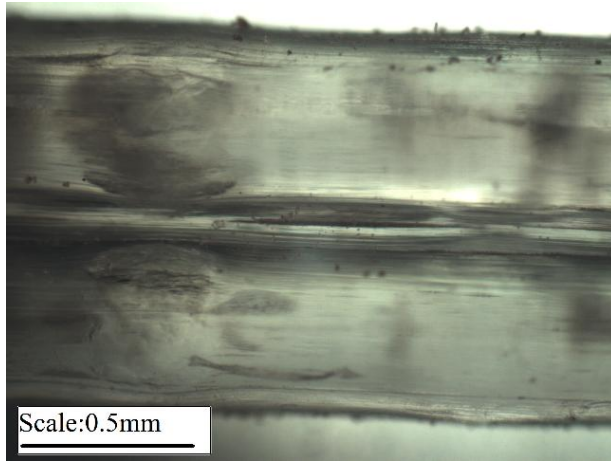


Debonding

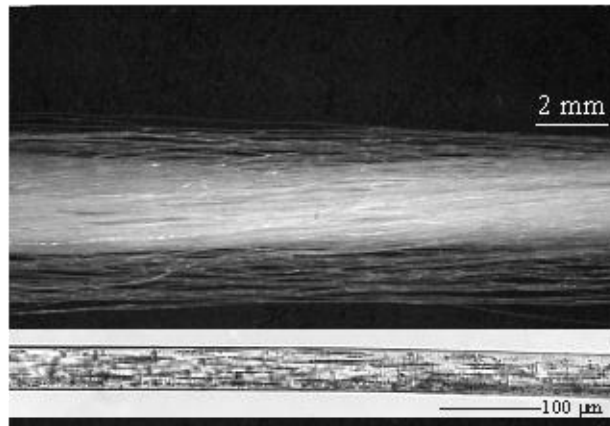


Frictional Sliding

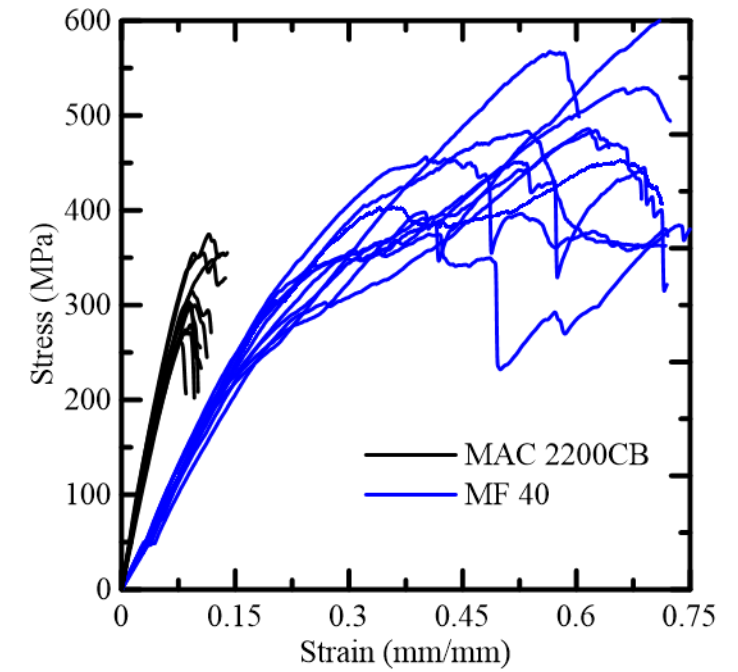
Interface Characterization PP based Yarn/Fiber and Textile technology



1) Macro-synthetic fiber – MAC 2200CB
Chemically enhanced macro-fiber



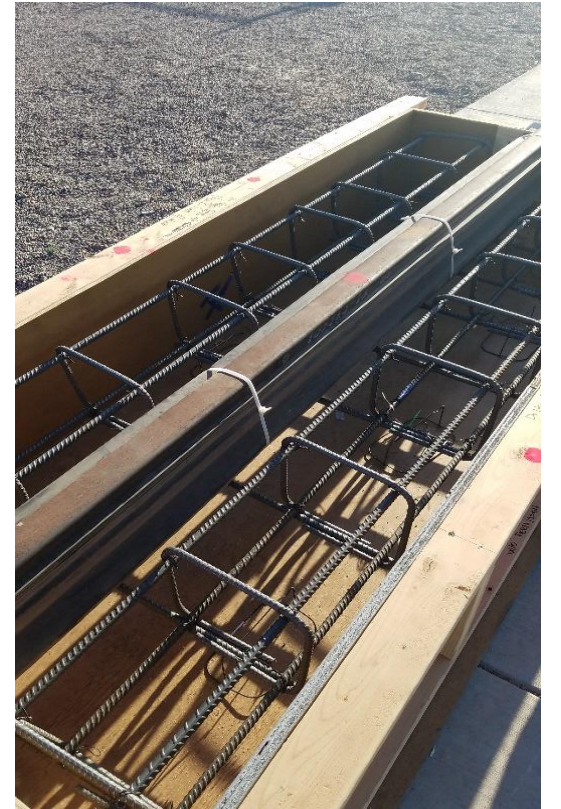
2) Microfiber –Fibrillated multi filament micro-fiber
500 filaments of 40 microns per yarn



	MAC 2200CB	MF 40
Loading Rate (mm/min)	0.4	2.5
Gage Length (mm)	25	25
Effective Yarn Dia. (mm)	0.82	0.89
Tensile Strength (MPa)	311 (+/-38)	492 (+/-65)
Elastic Modulus (MPa)	4499 (+/-351)	1601 (+/-117)
Toughness (MPa)	34 (+/-12) MPa	5058 (+/-1748)

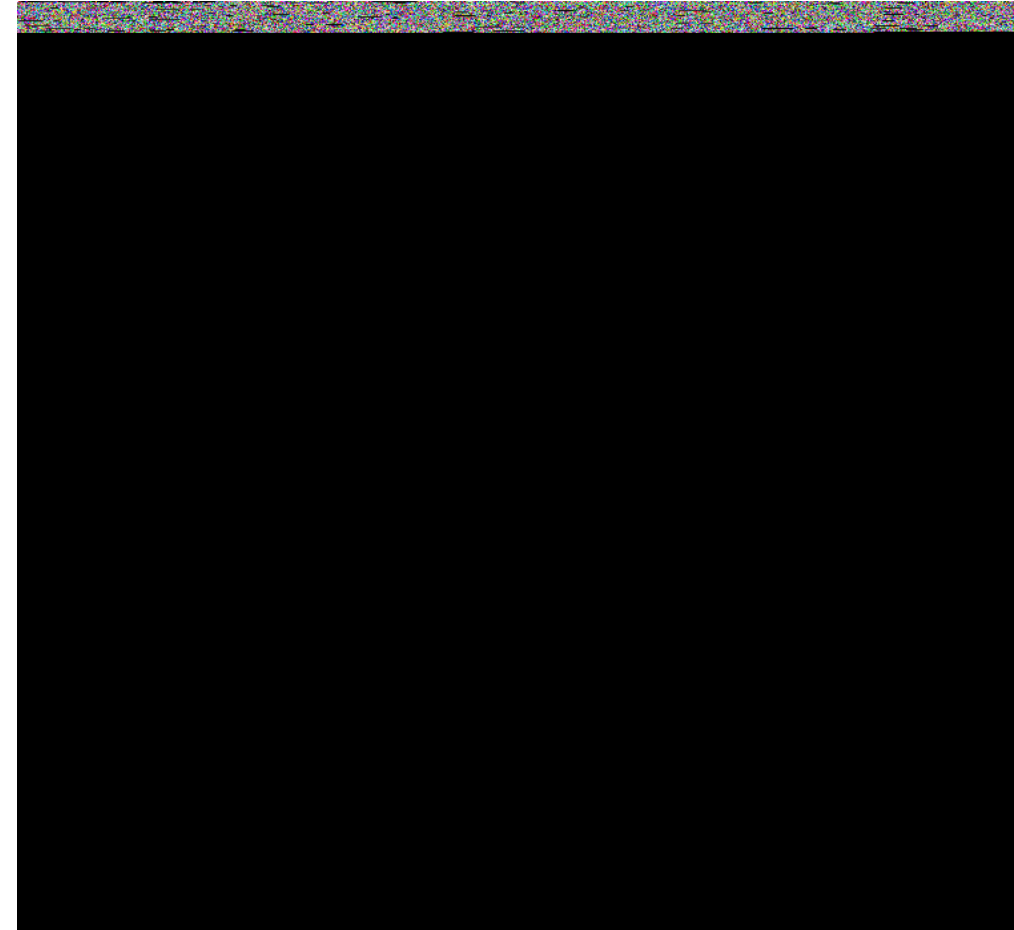
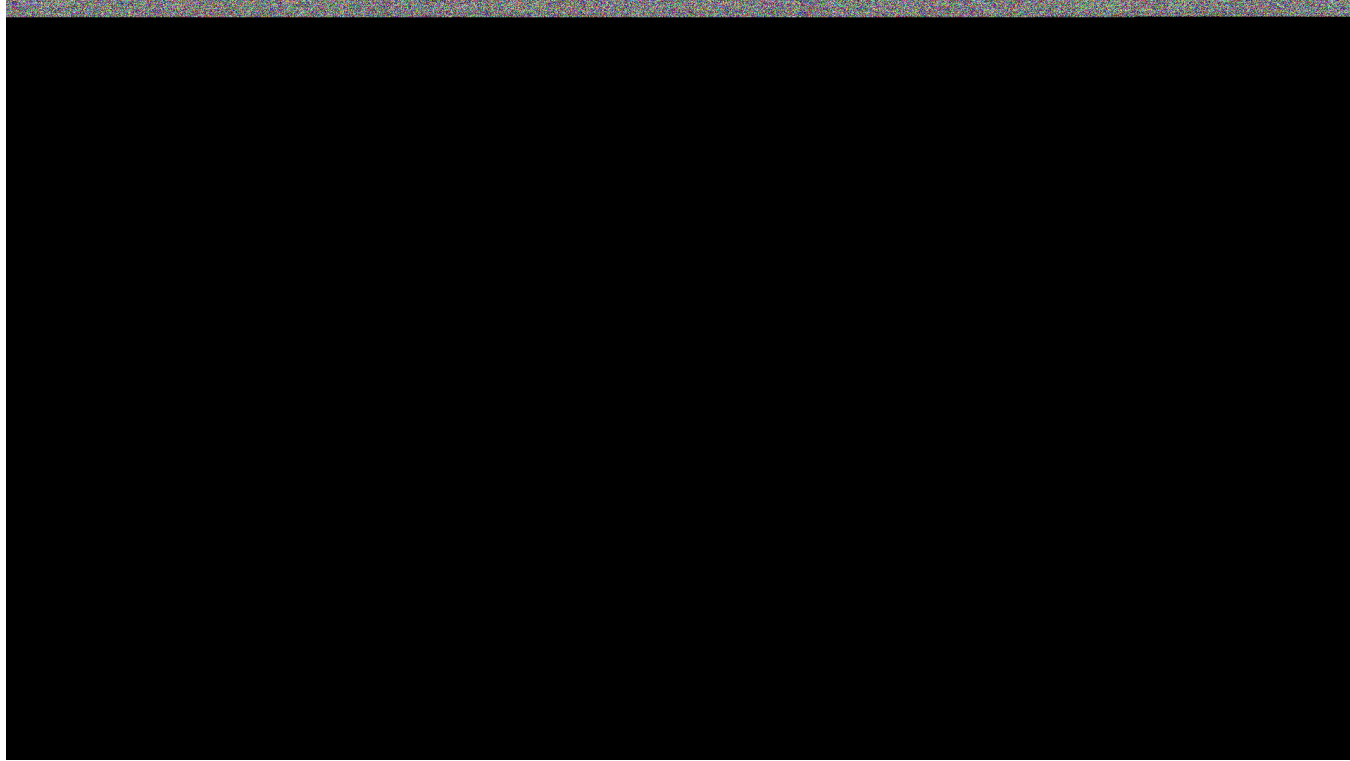
Light Rail in the City of Phoenix

Rebar utilization consumes significant time, Labor, durability

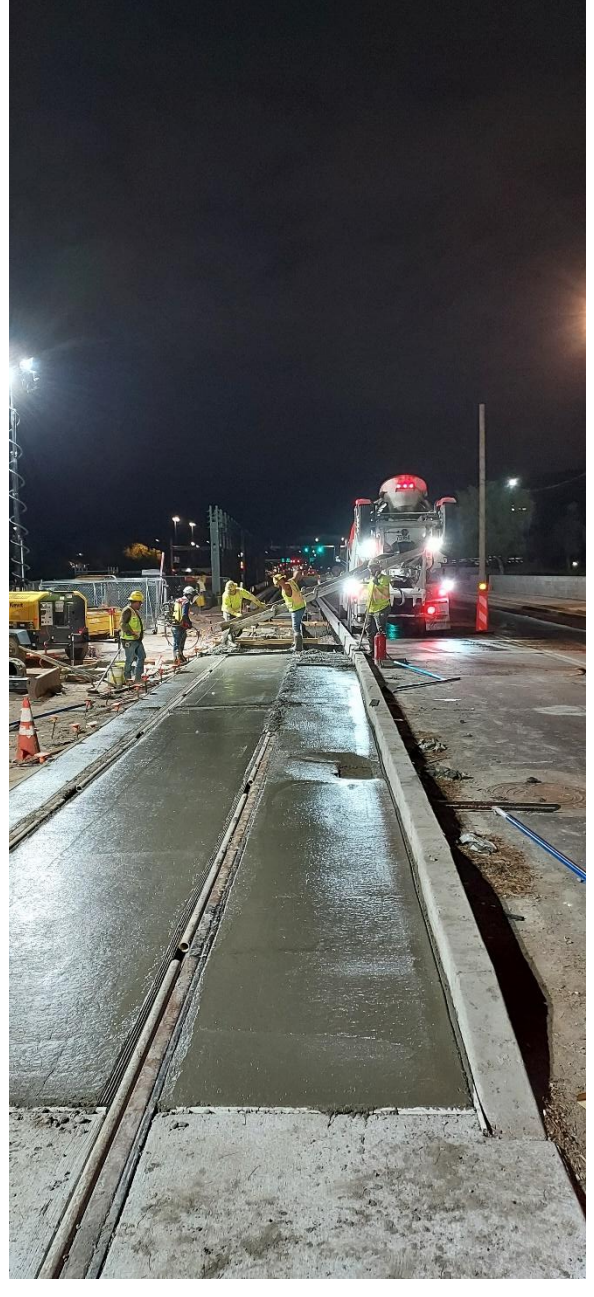
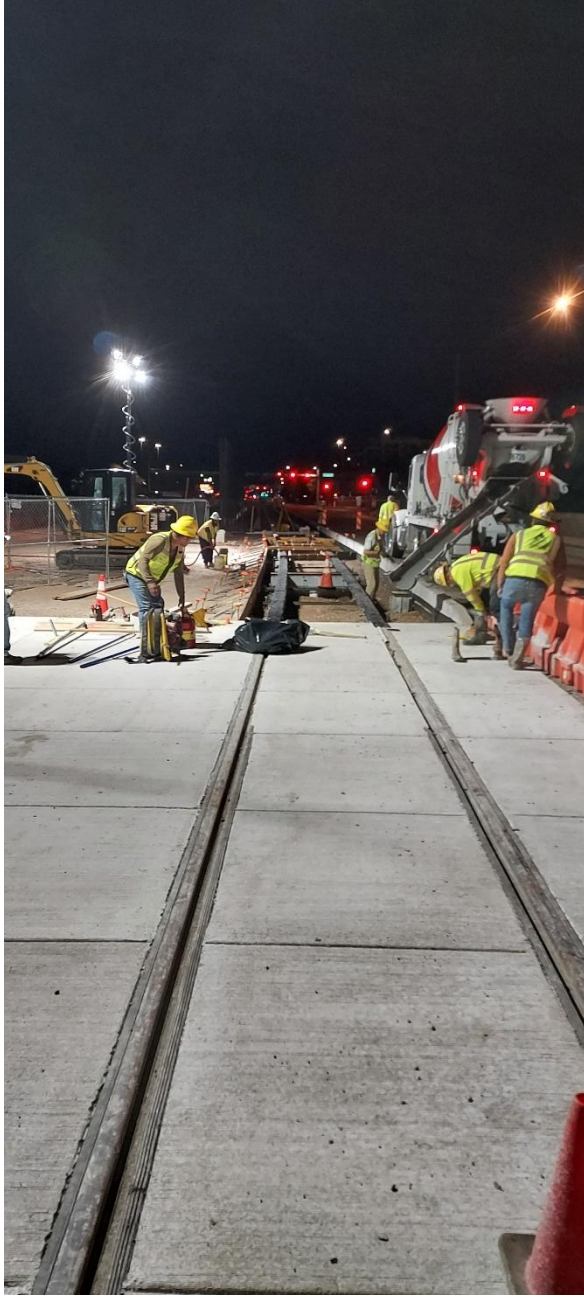
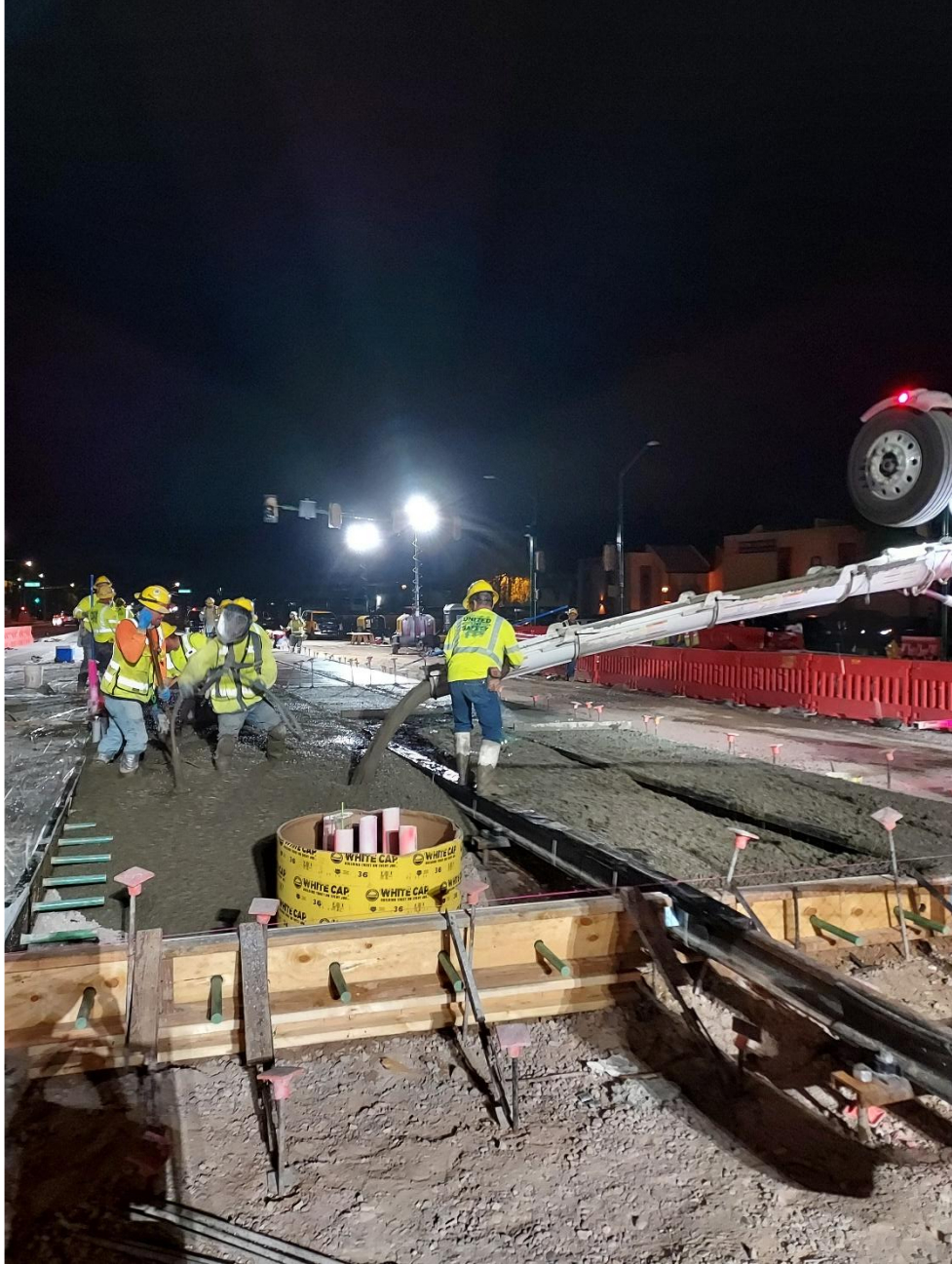


Fibers expedite the time and reduce cost

Full scale mockup Fatigue Testing (2 Million Cycles , 4Hz)



- Loading : 17.5 kips
- Data Acquisition of 100 cycles per 10,000 cycles, 2.0 million cycles, 8 days at 4 Hz

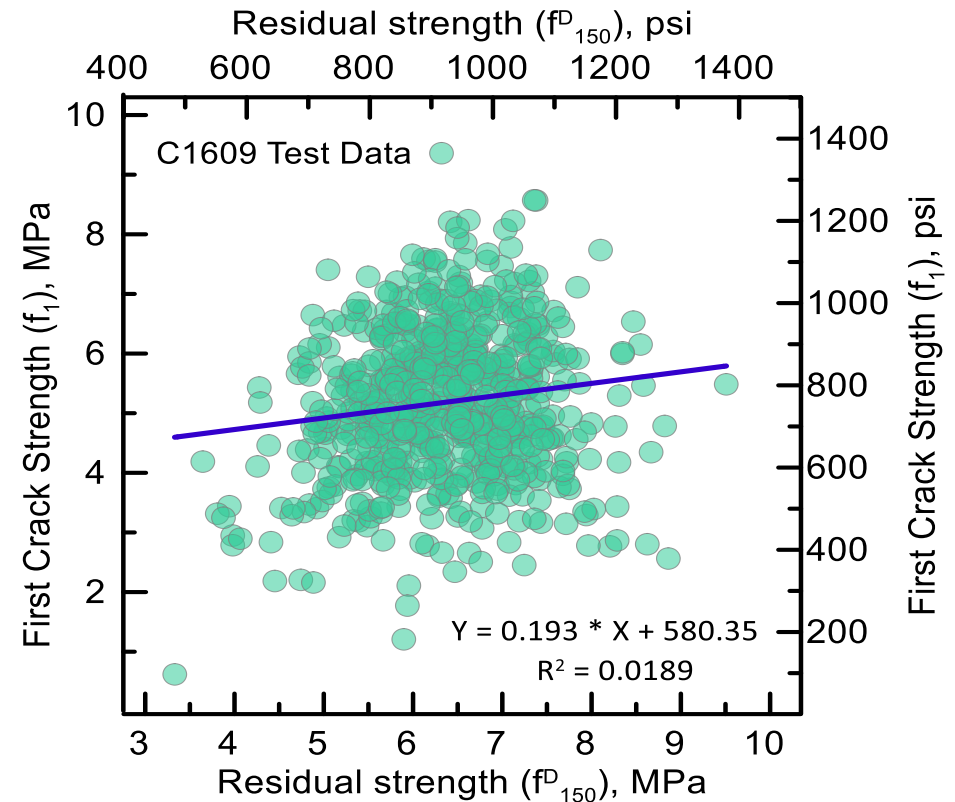
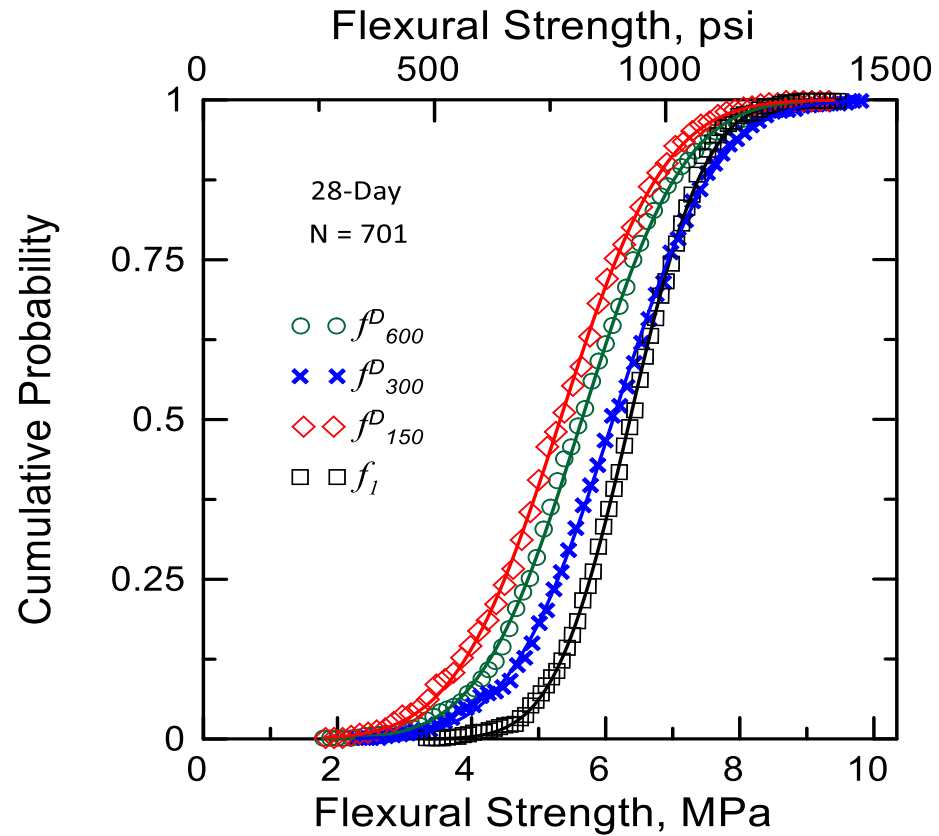


VERKA NASRI, DAVID KLUG, BRIAN FULCHER,
AND JAMES A. MORRISON

HANDBOOK OF PRECAST SEGMENTAL TUNNEL LINING SYSTEMS

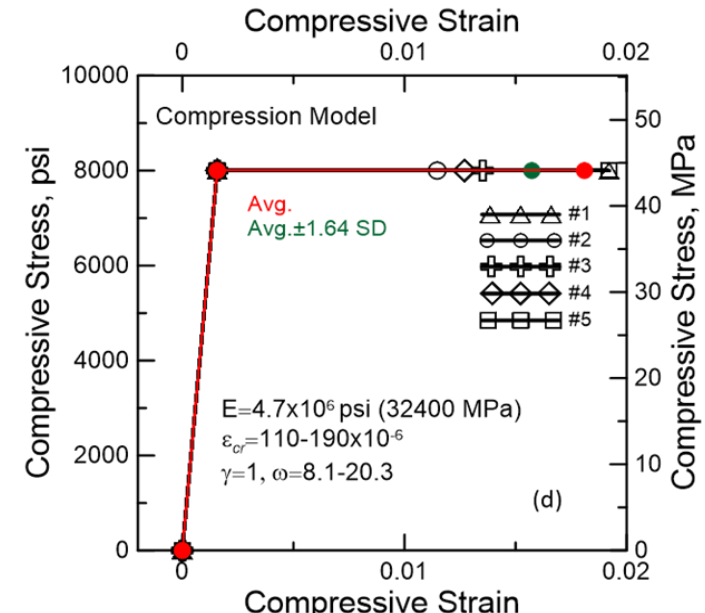
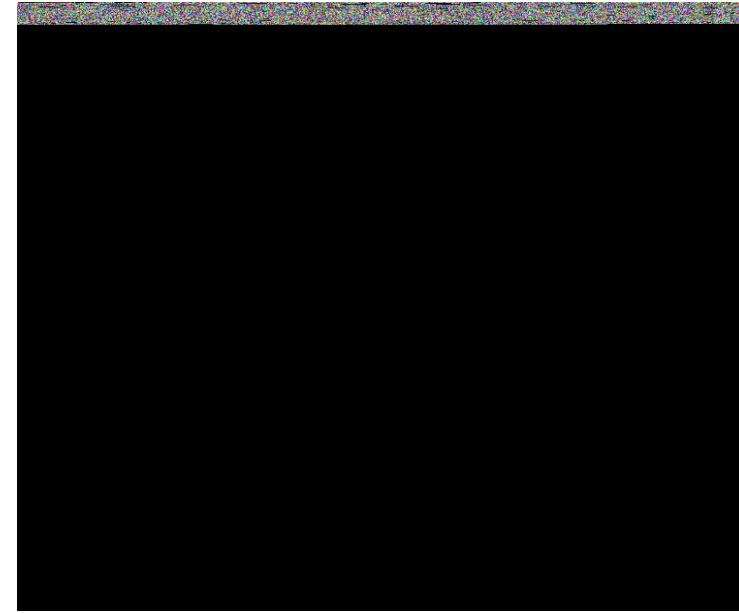
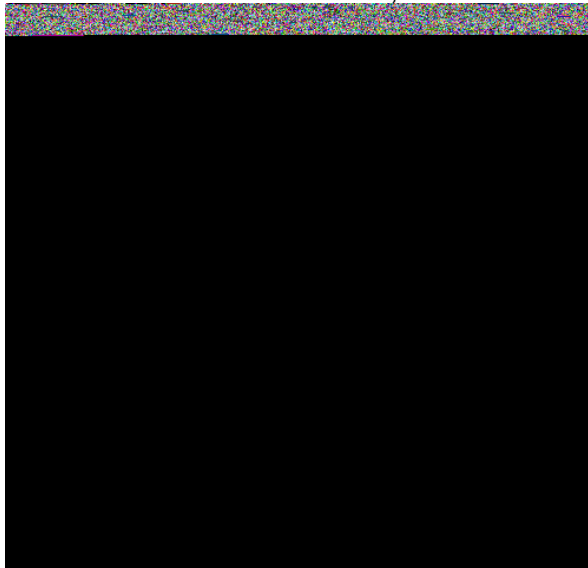
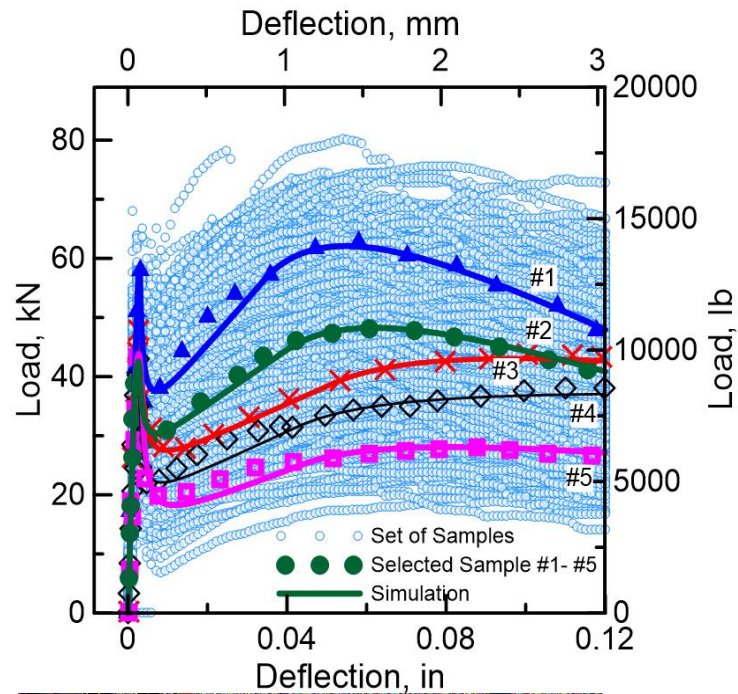


Statistical measures of distribution and correlation



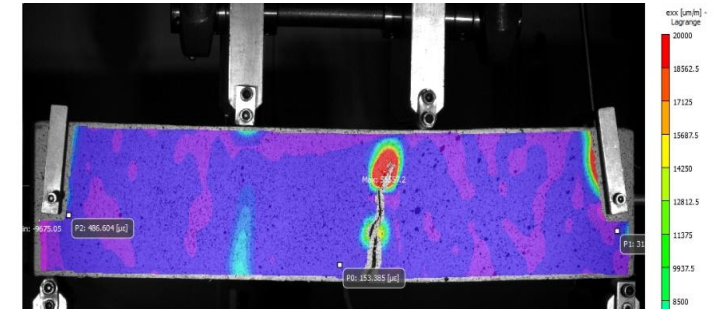
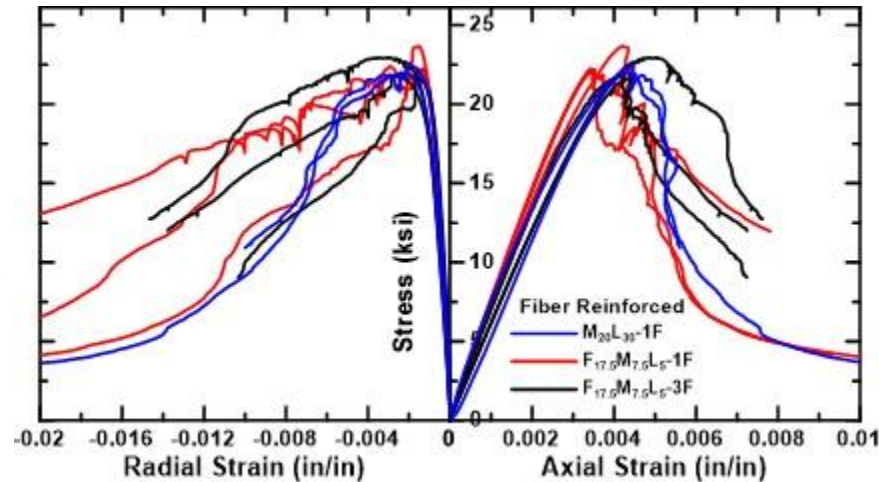
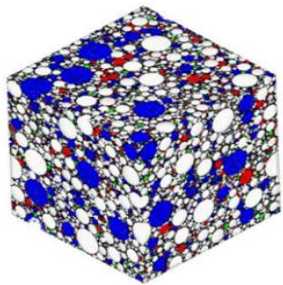
What do you do when there is no meaningful correlation?

Statistical measures of distribution and correlation, Simulation of QC data

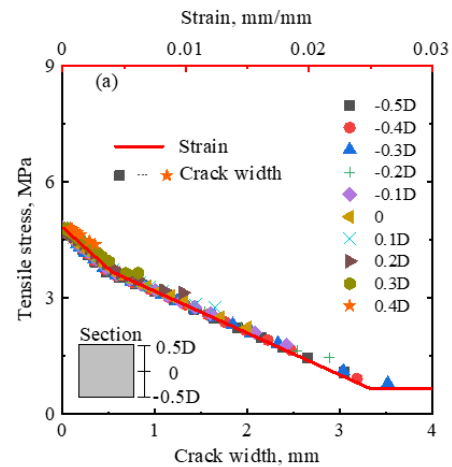
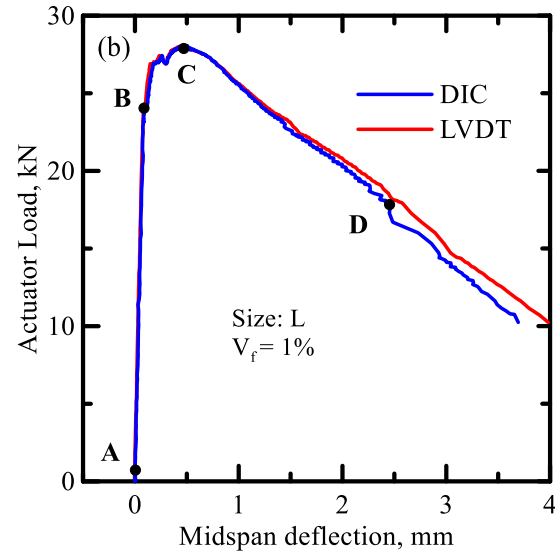
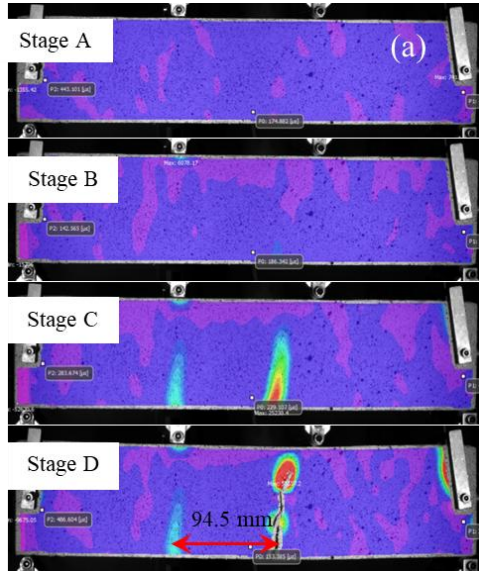
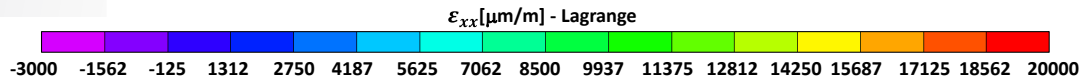


Non-Proprietary UHPC Concrete Mixtures for Bridge Rehabilitation Arizona, 2022-24

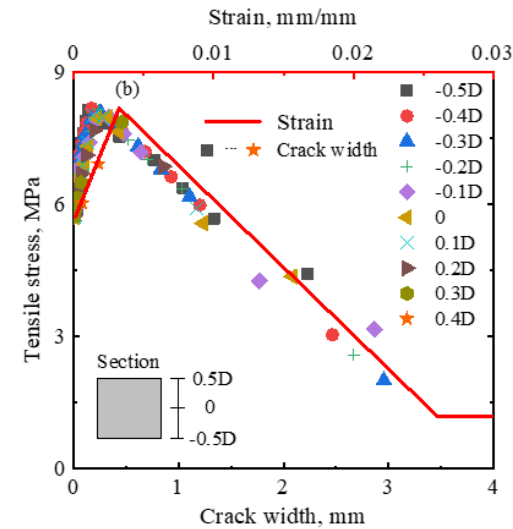
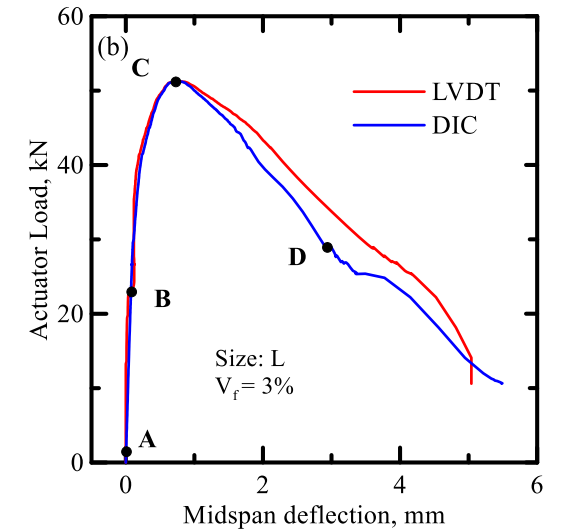
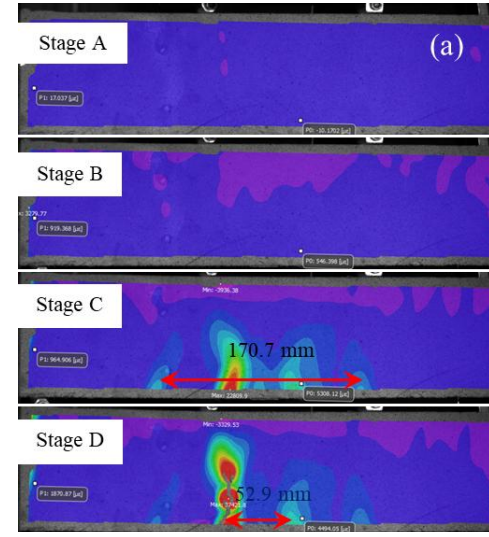
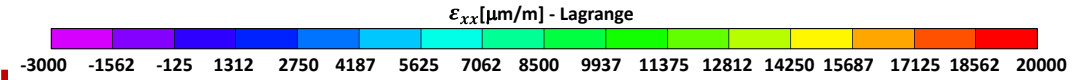
- Life cycle cost and economic advantages of using UHPC systems in terms of structural applications.
- Precast UHPC elements reduce operational complications, labor-intensive tasks. From a design perspective, elastoplastic behavior improves durability, ductility, load redistribution, and weight reduction.
- Develop Sustainable Materials, using solid mechanics, material formulations, structural components, and systems.
- solutions for composite materials for transportation, water treatment facilities, pipes, tunnel lining, thin sections, Structural Shapes



UHPC with 1% and 3% steel fibers, Stress Crack width Response

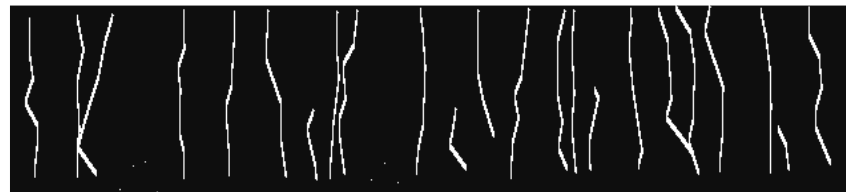
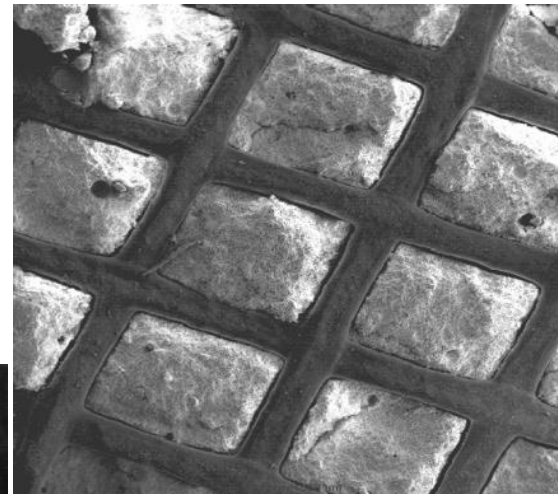
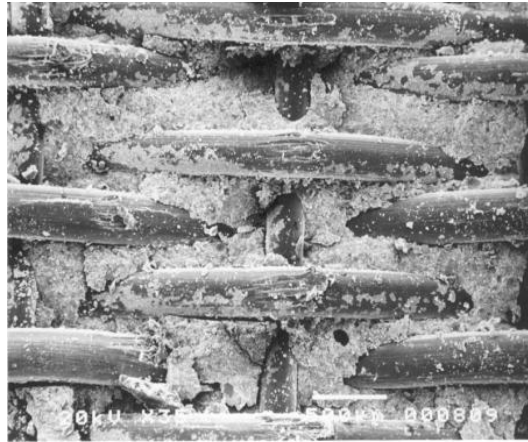
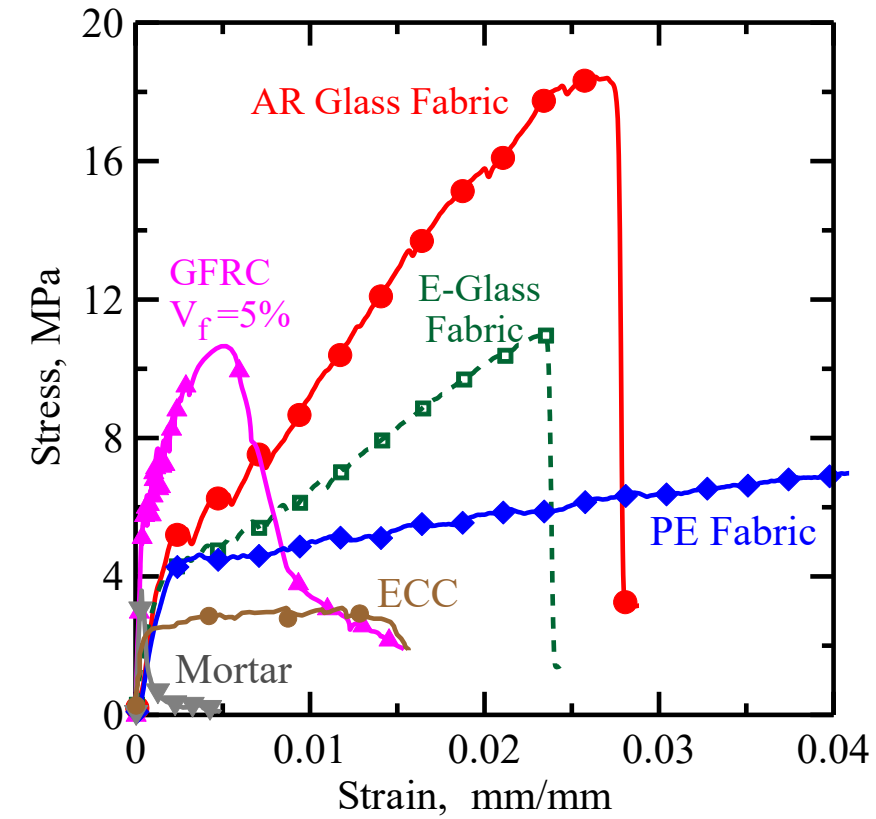


1% steel fibers



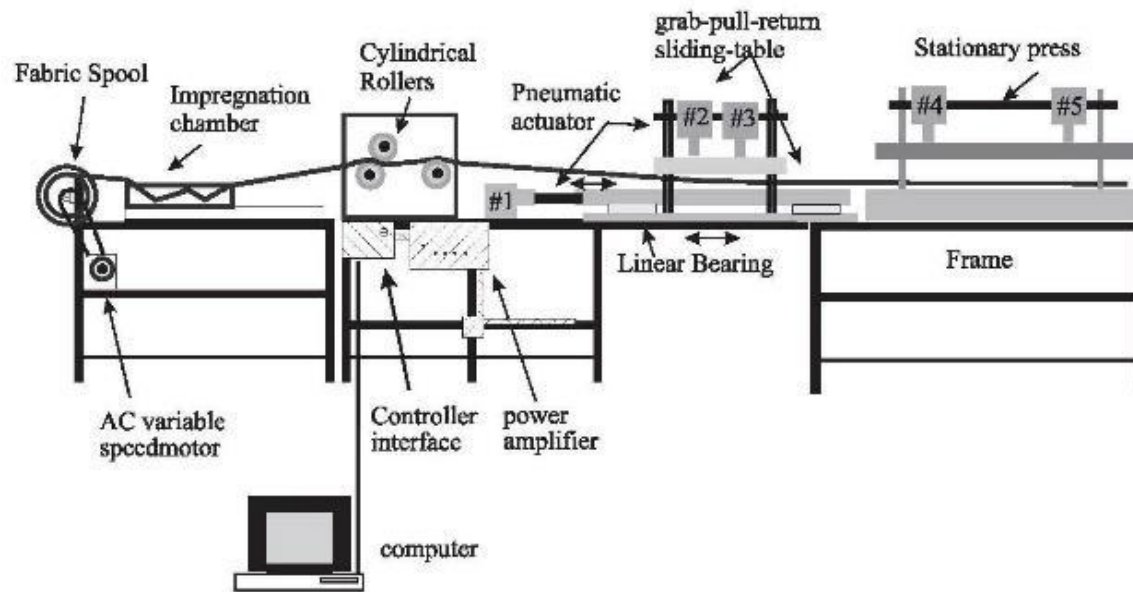
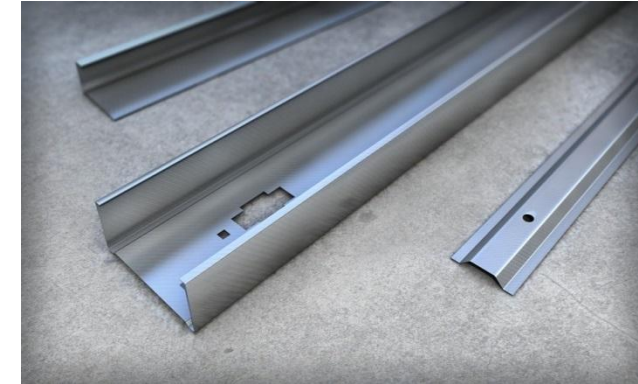
3% steel fibers

Textile Reinforced Concrete



Structural Shapes: Development, Analysis, and Implementation using Design Approach

Automated pultrusion system, full scale structural shapes composed of TRC laminates can be manufactured efficiently and effectively.

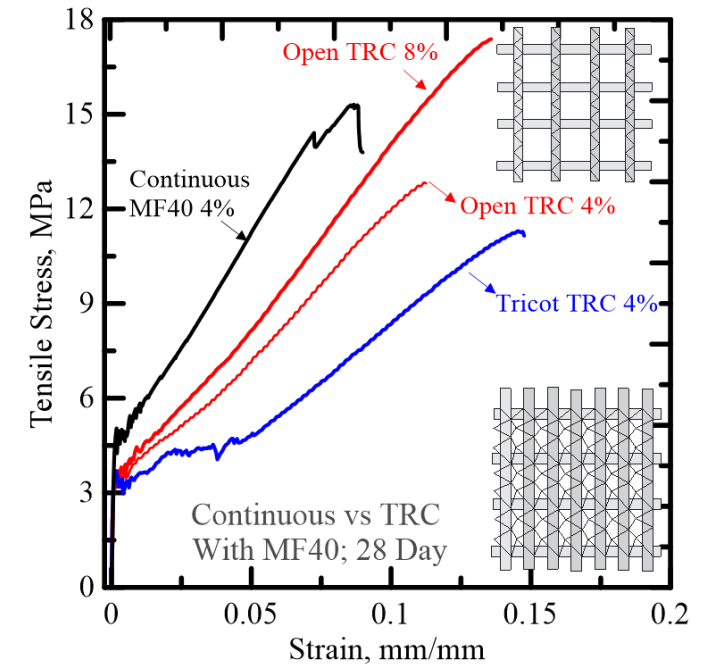
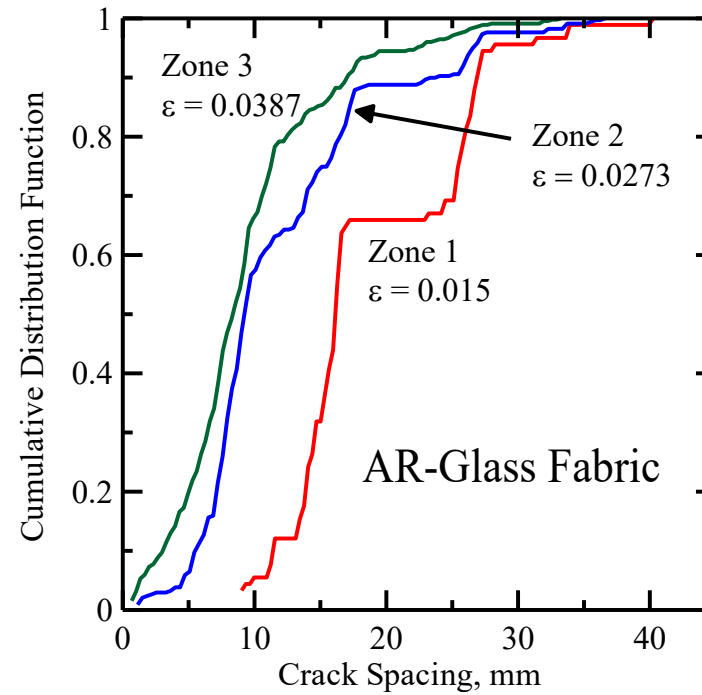
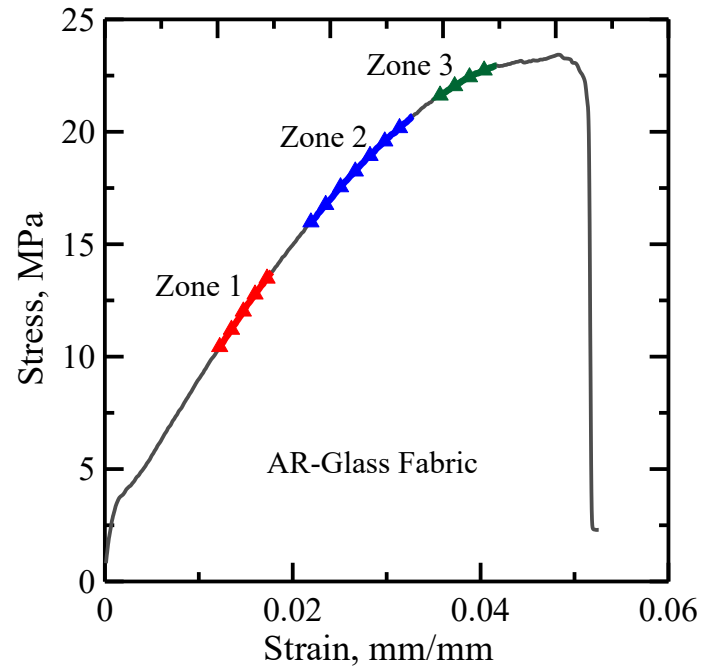


Pultrusion Process Schematic Diagram

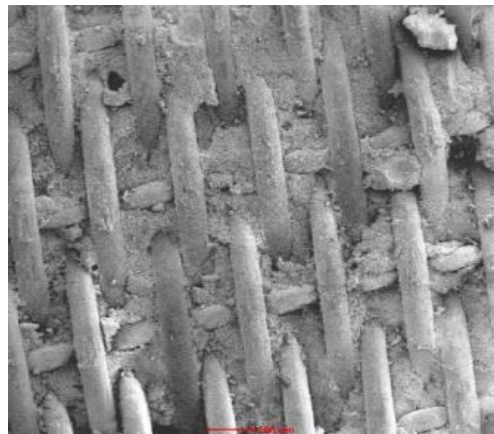


Light gage steel sections

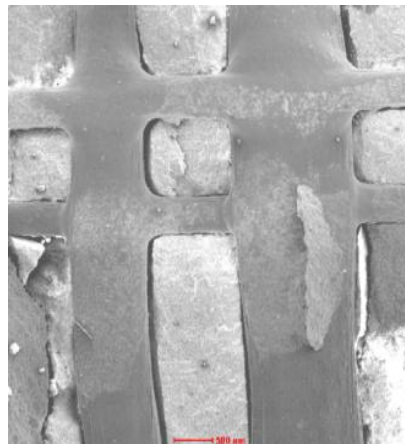
Homogenization of Crack spacing –Mechanical interlock



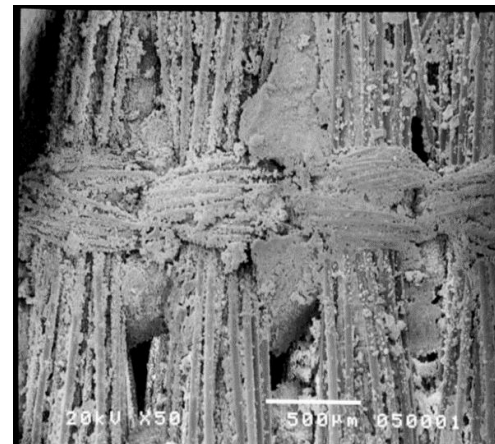
Polyethylene (PE) Woven



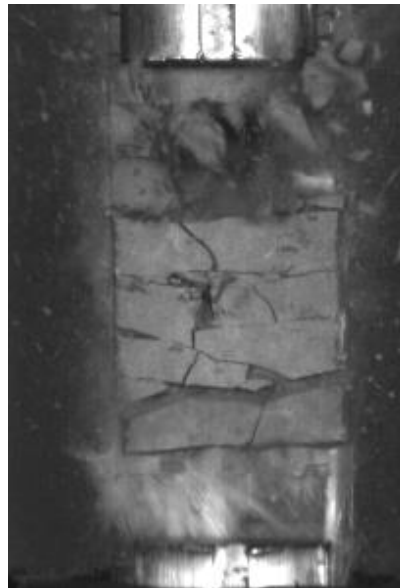
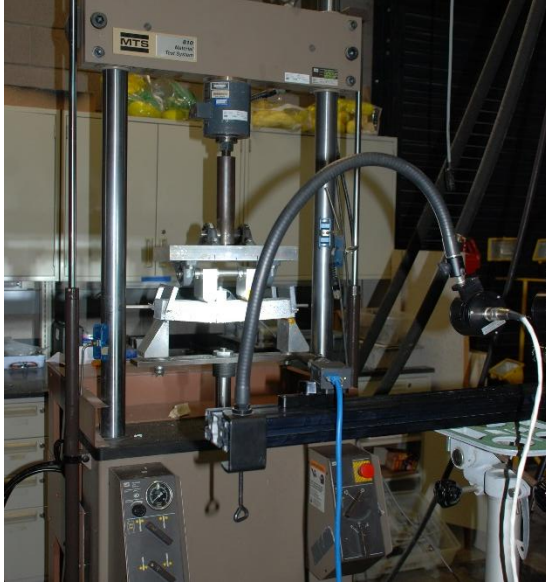
AR Glass Bonded



Polypropylene (PP) Knitted

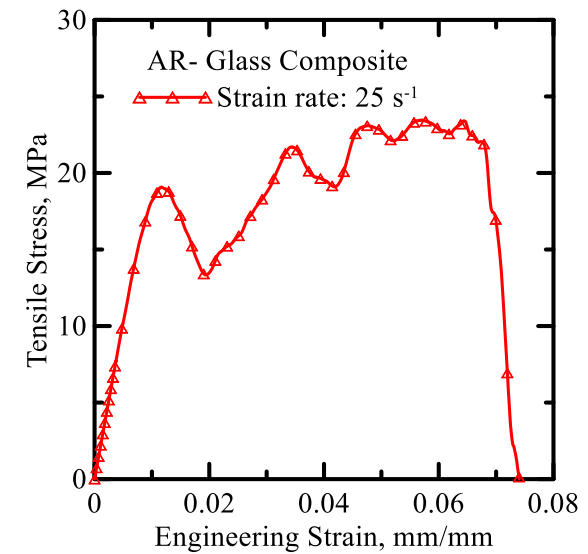
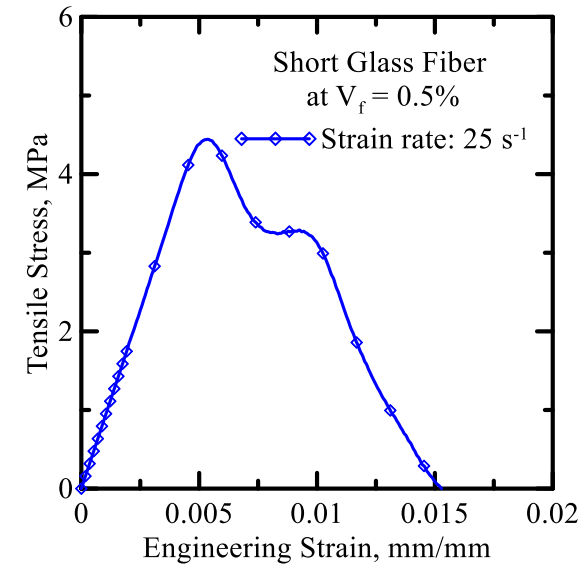
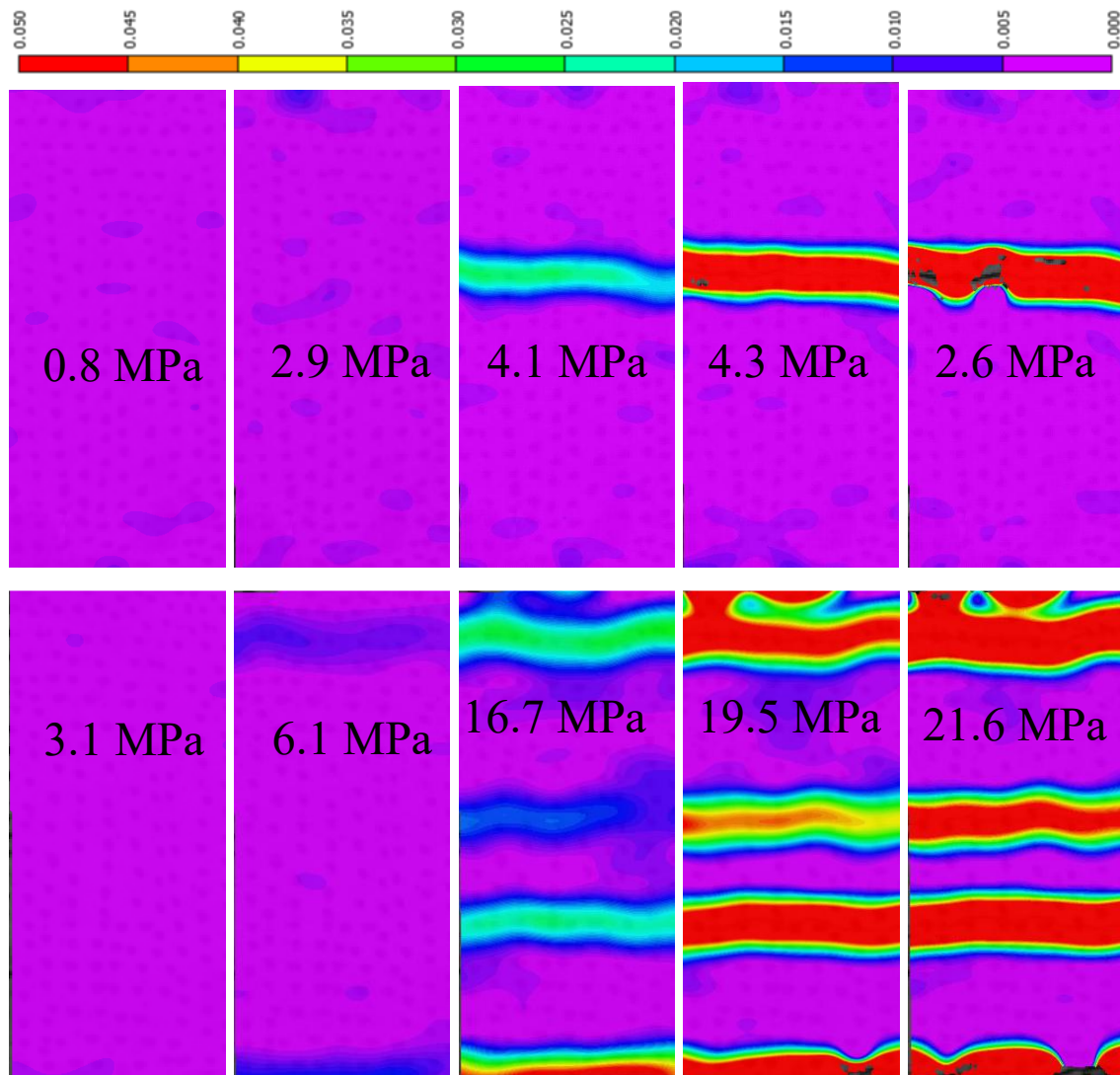


Constitutive modeling under Static, Impact, and High-Speed Testing under Tension and Flexure

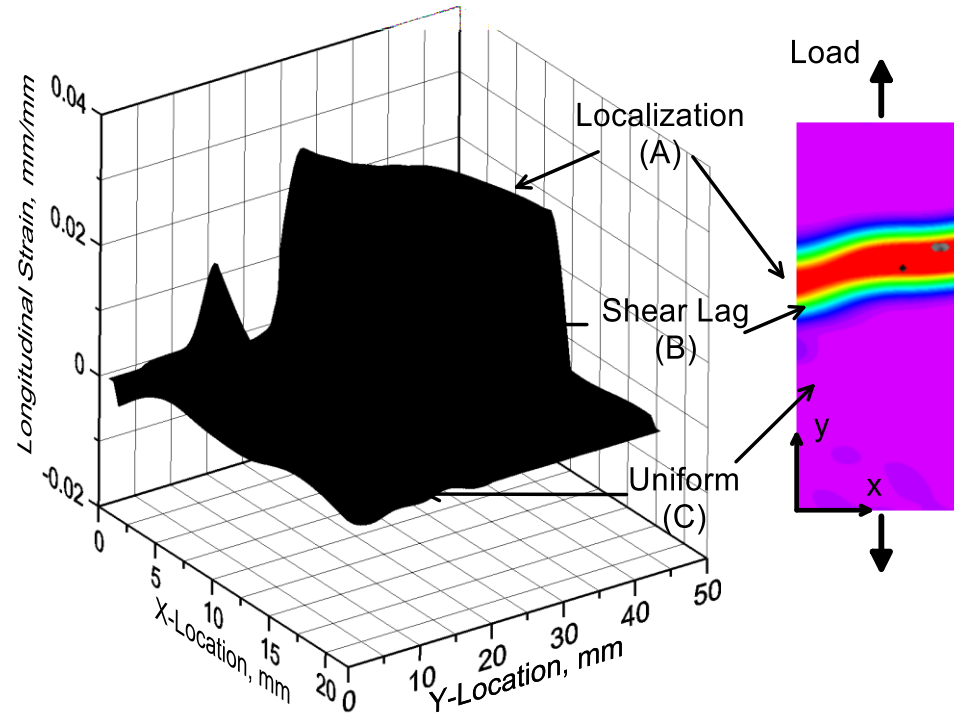


Strain Map of Tension Stiffening

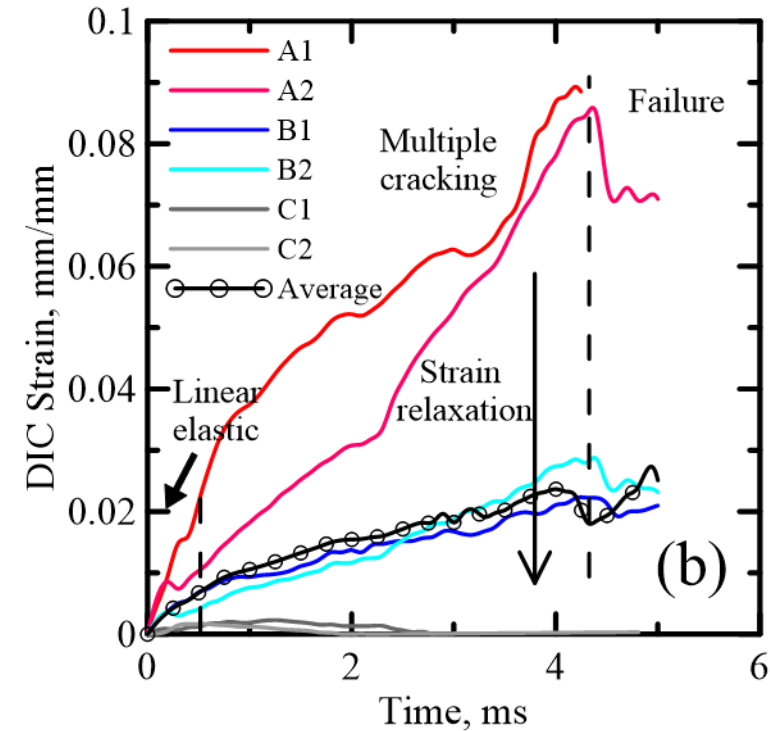
Short fiber vs. continuous fiber systems



Quantification of DIC strain



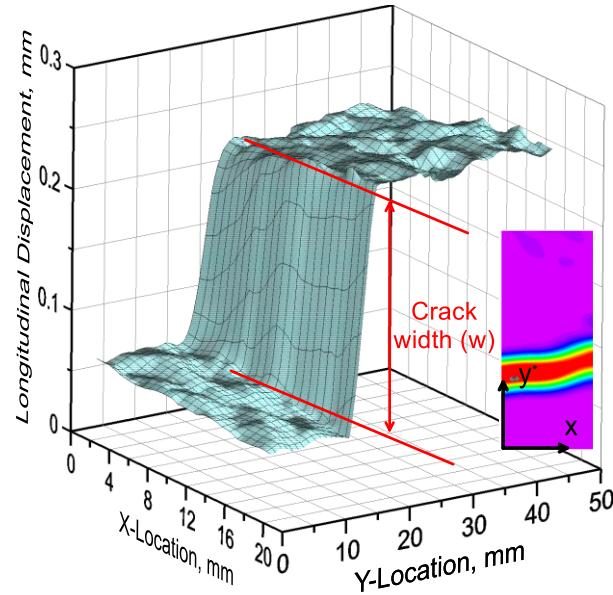
Identification and label of each zone



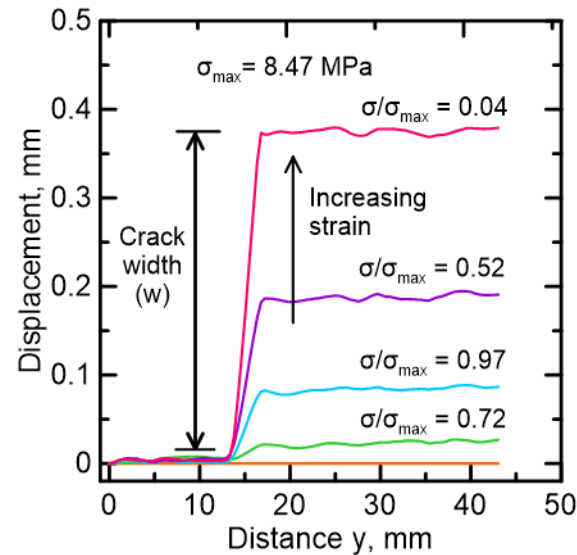
DIC strain versus time histories at different zone

- A: Localization Zone – Fiber debonding
- B: Shear Lag Zone – Shear lag bonding stress distribution
- C: Uniform Zone – Fiber and matrix are perfectly bonded

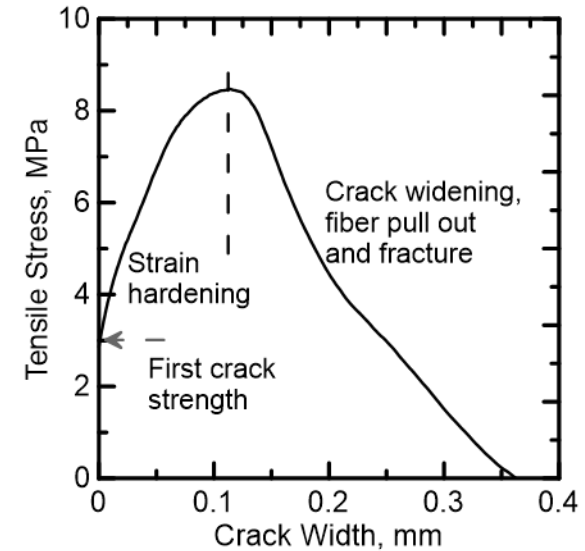
Crack Width Measurement



Displacement Field



Displacement
Distribution Along
Specimen

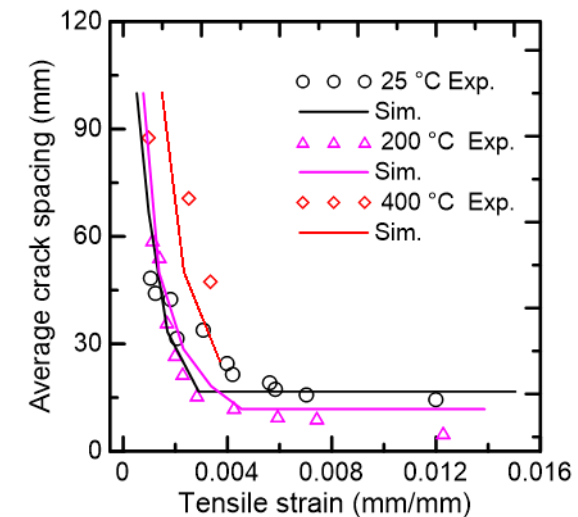
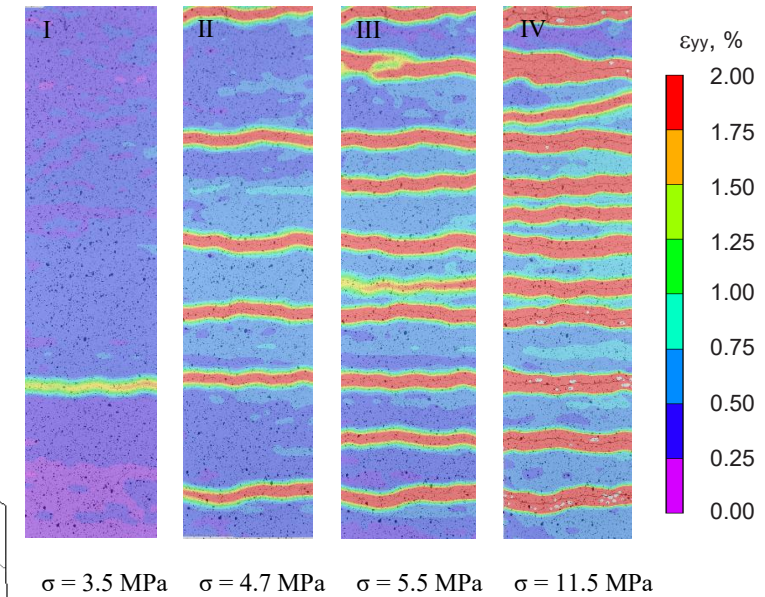
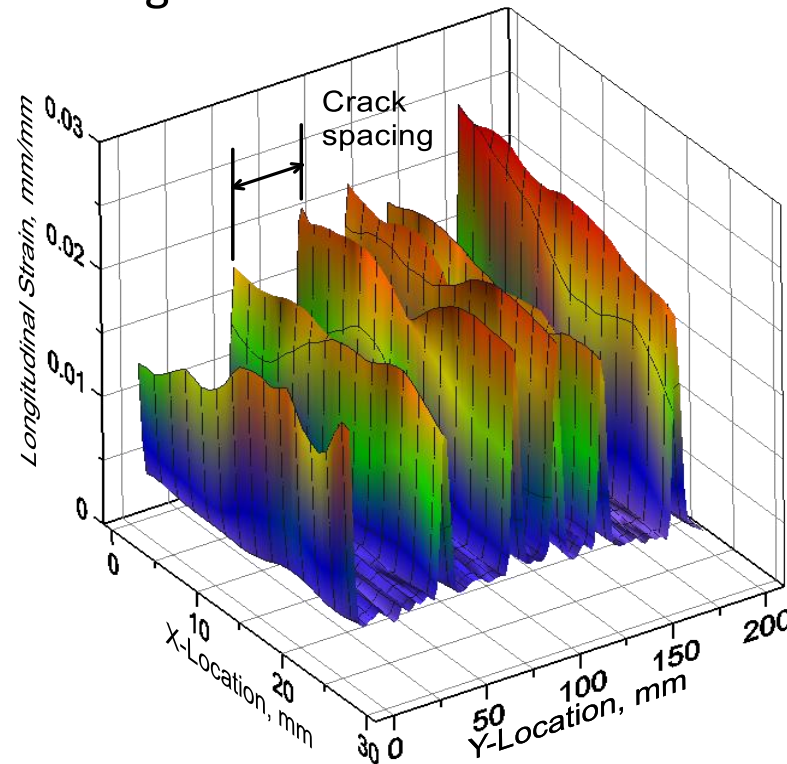


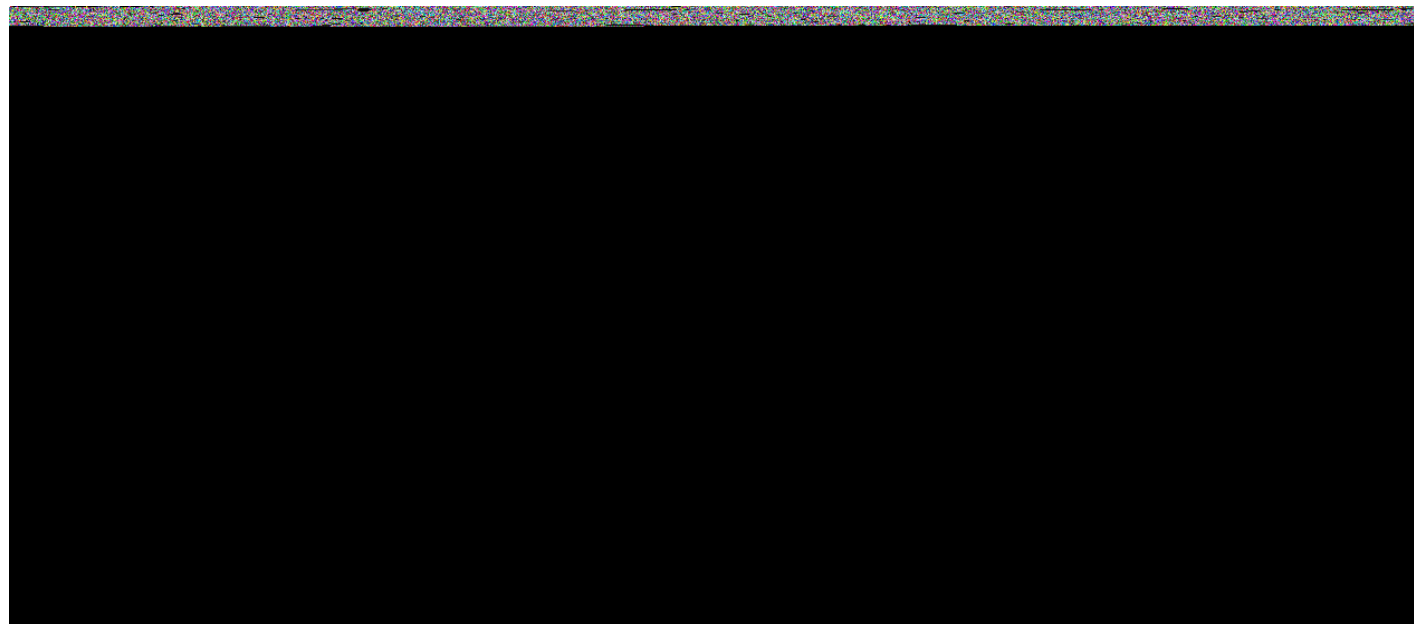
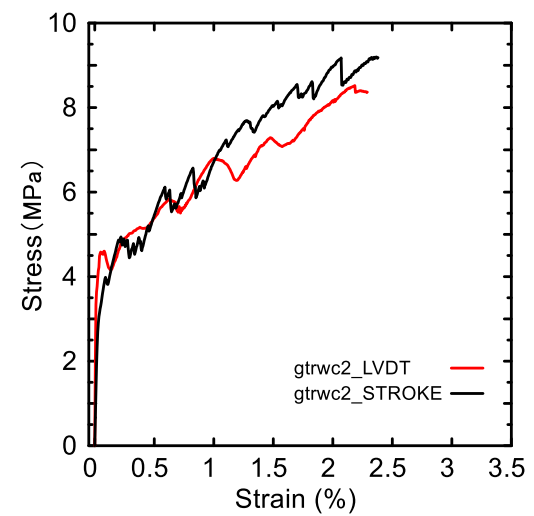
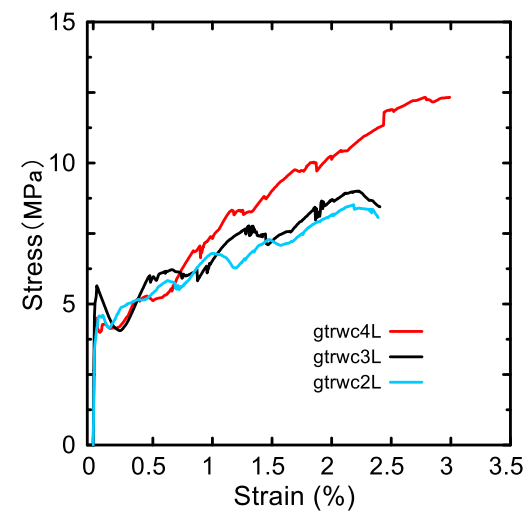
Stress-Crack Width
Relationship

- Non-contact measurement
- Quasi-static to high speed
- Single crack and multiple cracks

Evolution of Crack Spacing in TRC

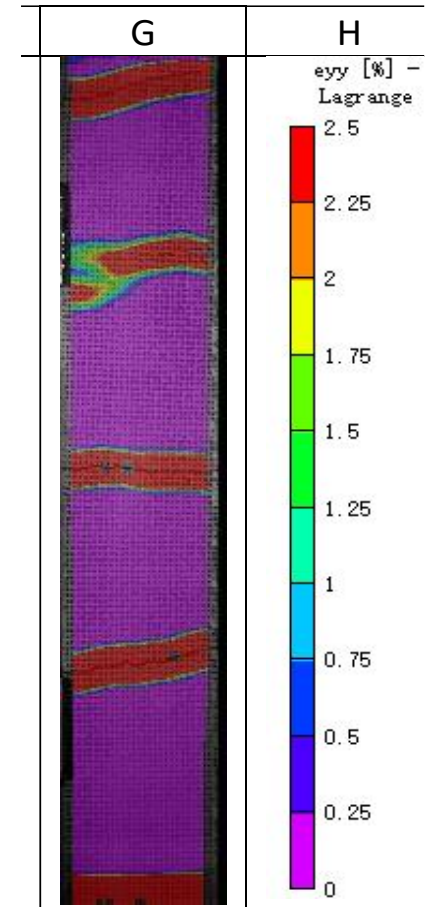
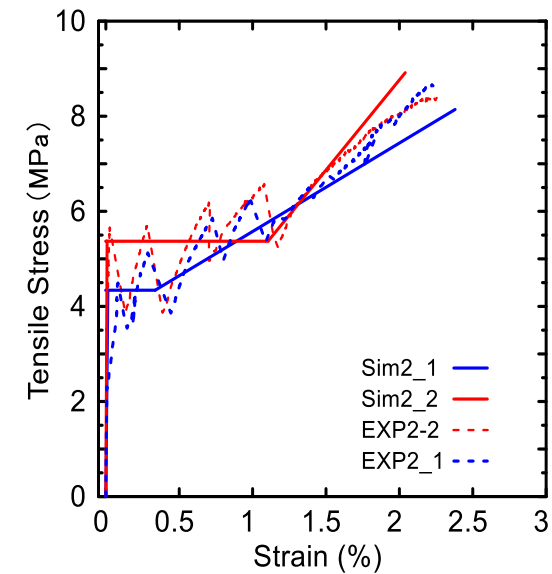
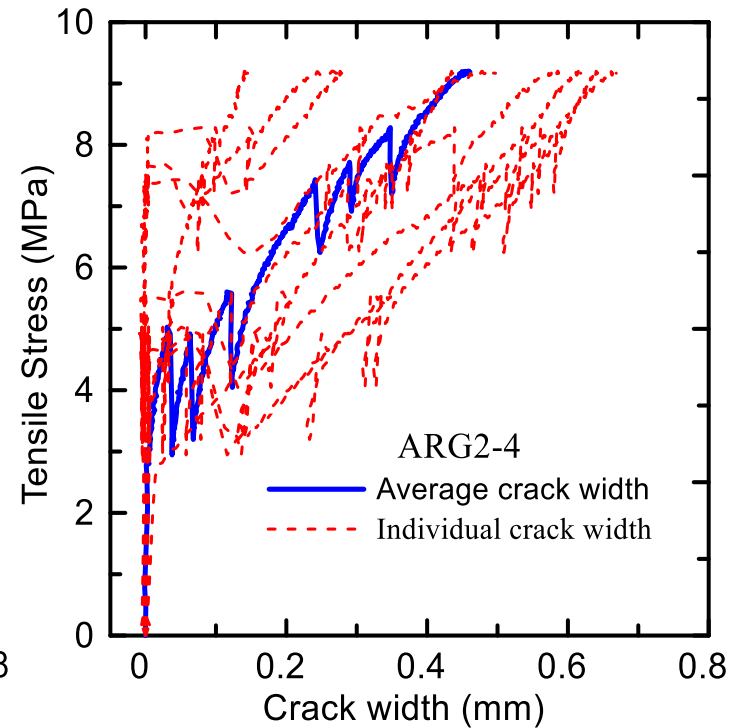
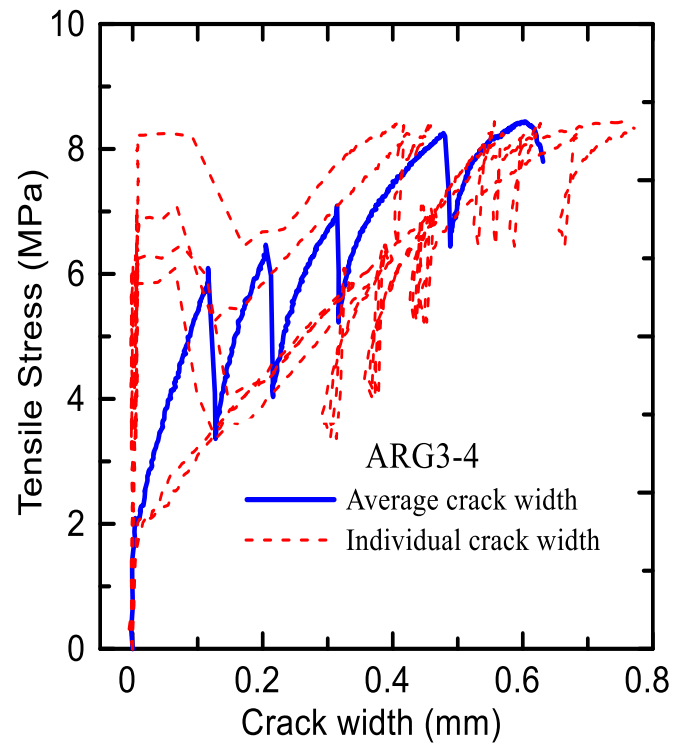
- Multiple cracking in tension
- Tension stiffening
- Development of parallel cracks
- Indication of toughening mechanisms
- Corresponding to the characteristic length in numerical modelling



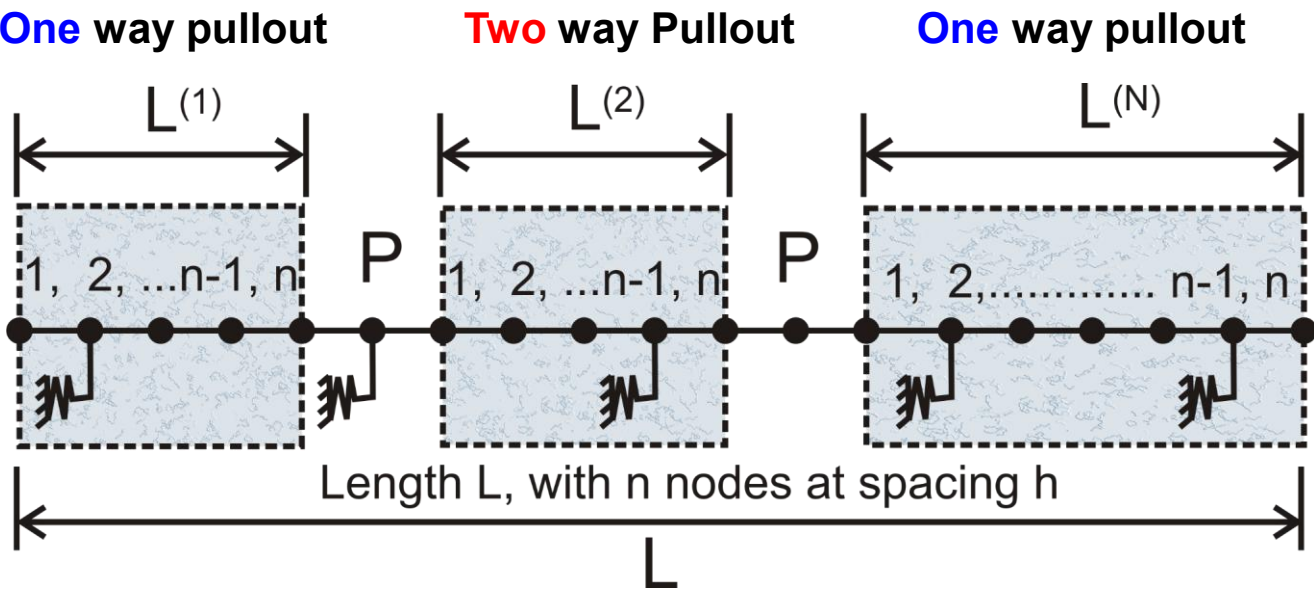
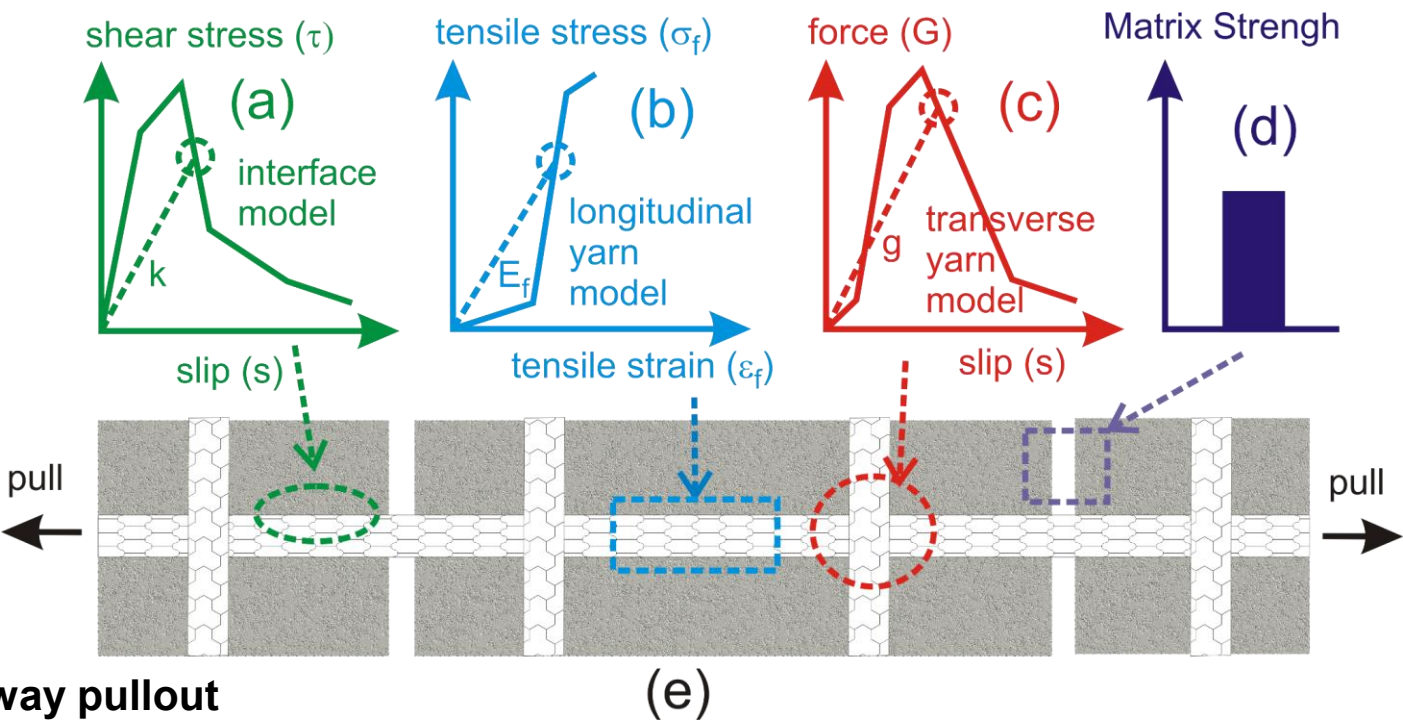


Crack Width Measurements

- Interacting Cracks enter a competition of sequential opening steps leading to incremental growth.
- There is no steady state unique crack growth process until localization dominates the response

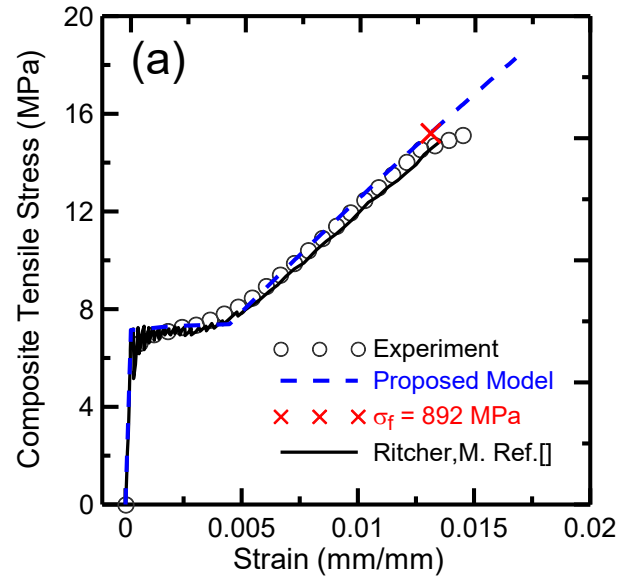


Modeling of Sequential Cracking

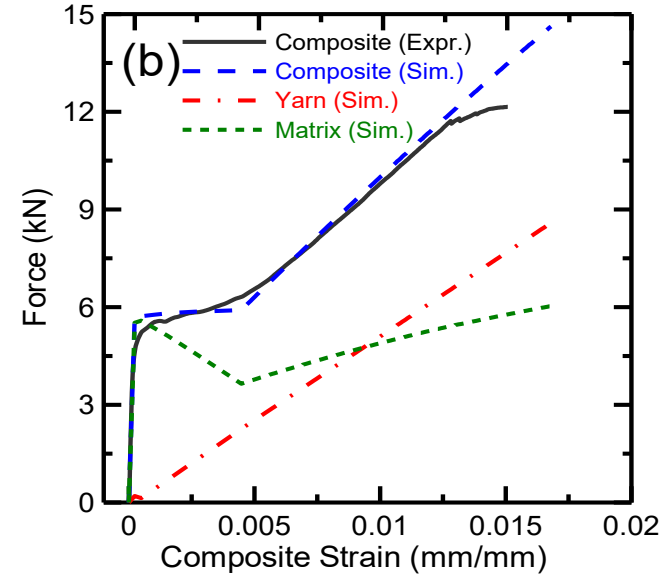


Tension Stiffening Results

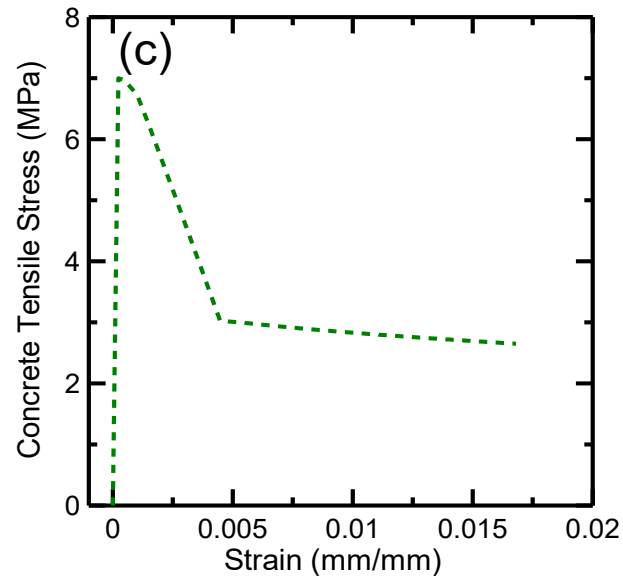
**Tensile stress
strain response**



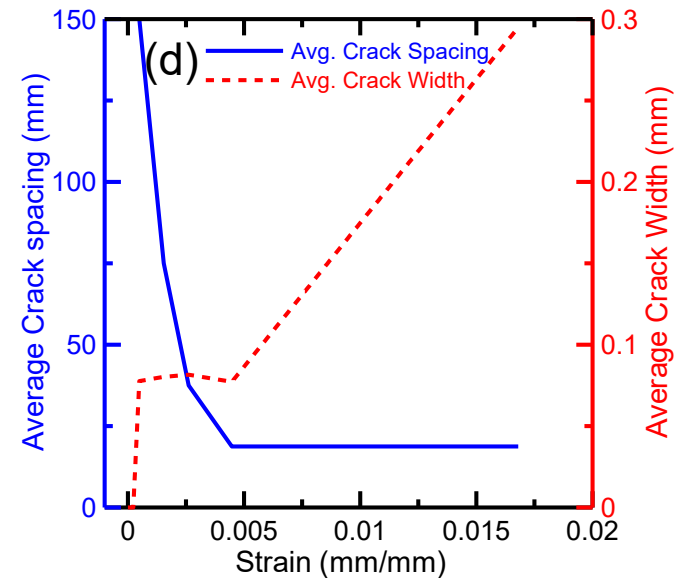
**Each force
component**



**Tensile stiffening
effect in concrete**

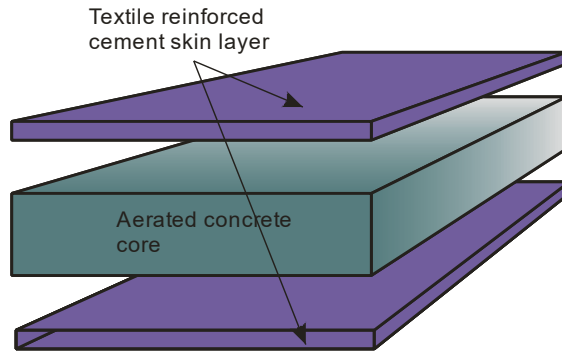


**Averaged
Crack
spacing &
crack
width**

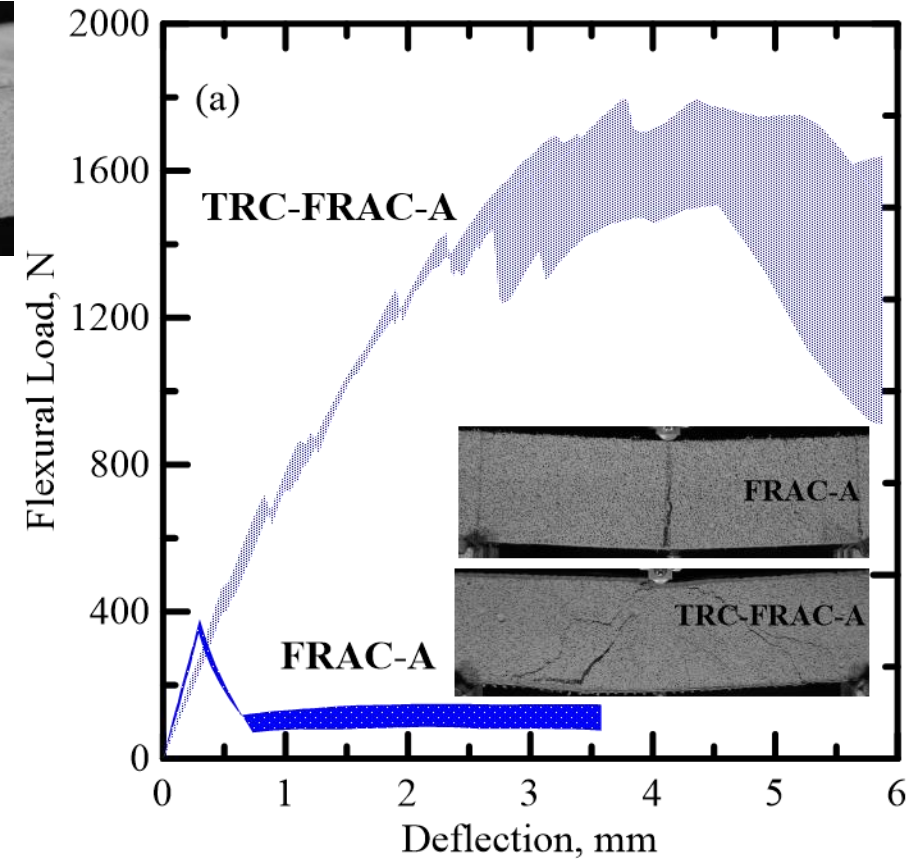
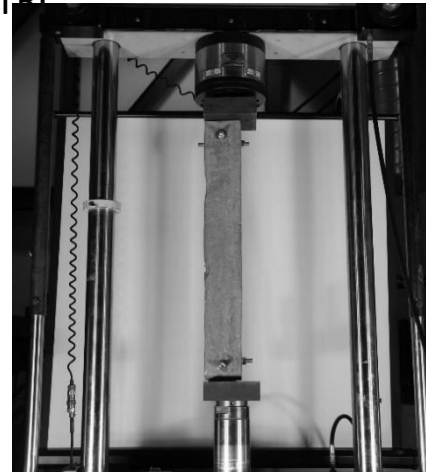


Sandwich Composites TRC Skin-AAC core

- Sandwich composite systems with TRC and light-weight aerated concrete core

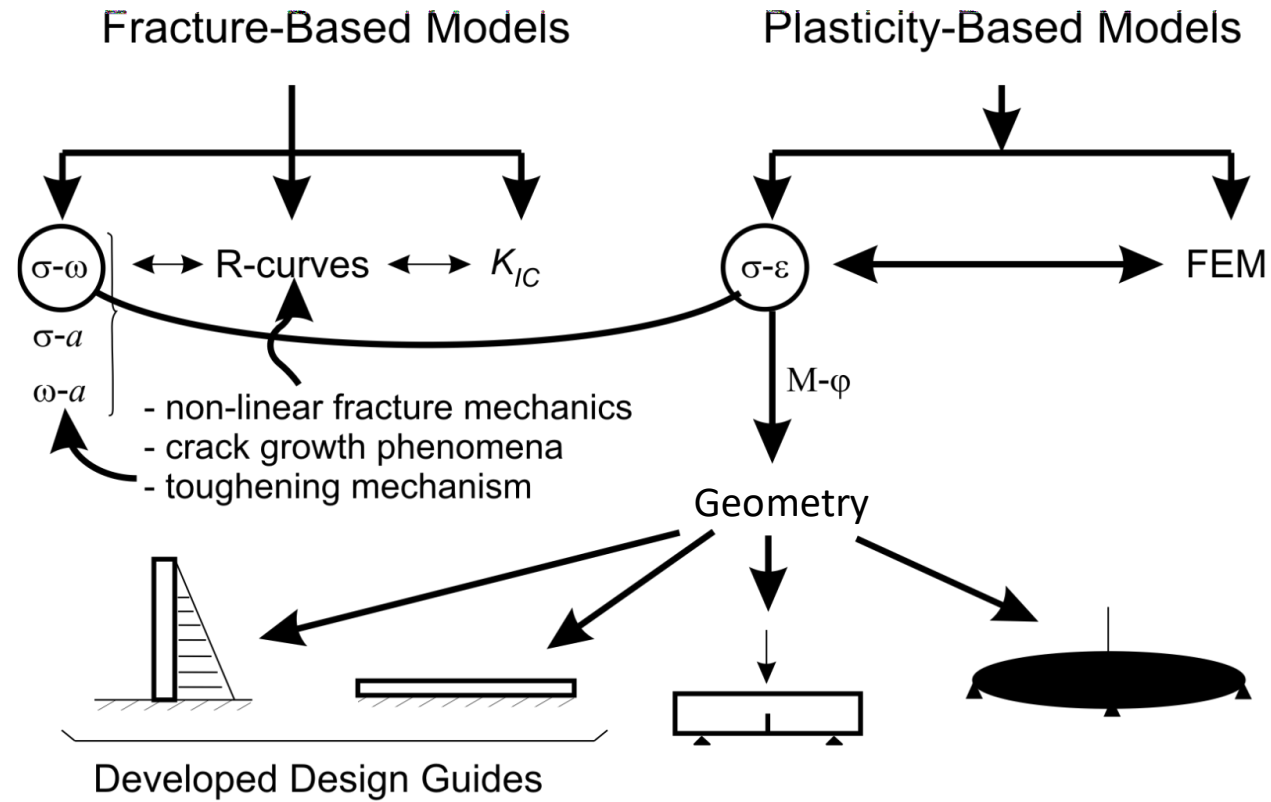


- Structural sections with TRC



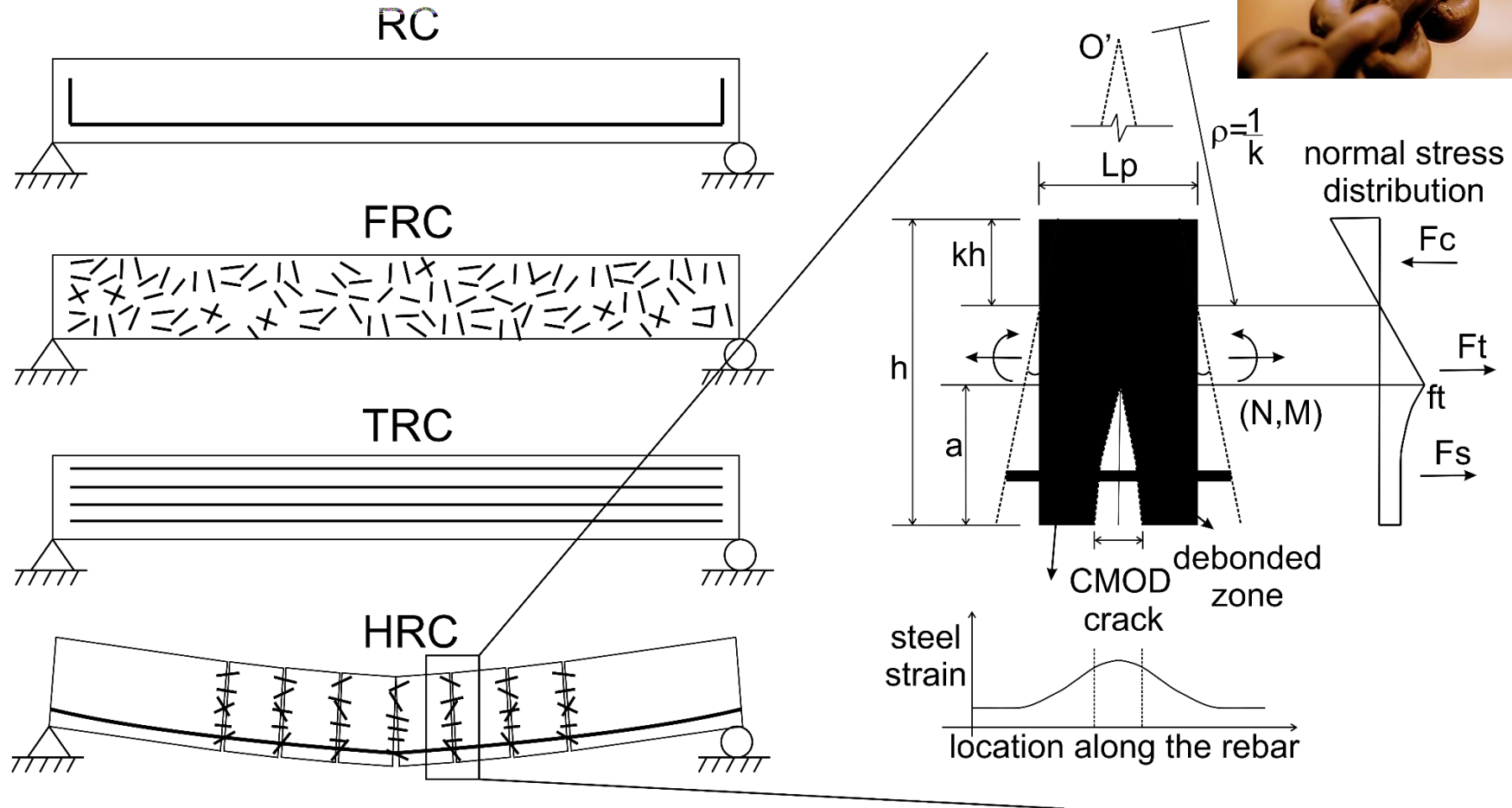
Dey, V., Zani, G., Colombo, M., Di Prisco, M., Mobasher, B., "Flexural Impact Response of Textile-Reinforced Aerated Concrete Sandwich Panels", *Journal of Materials and Design*, 2015, doi: 10.1016/j.matdes.2015.07.004

Fracture and plasticity models



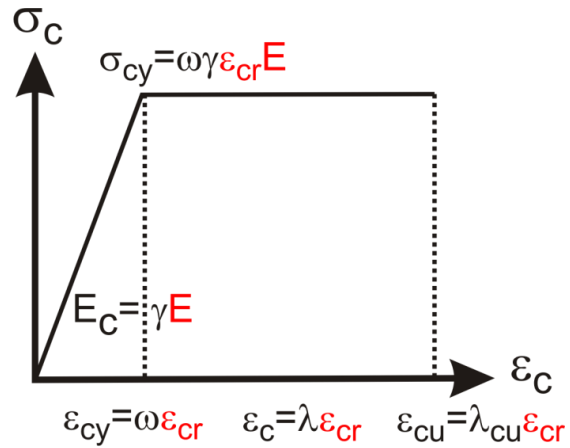
Nonlinear hinge

RVE in the context of a nonlinear hinge framework

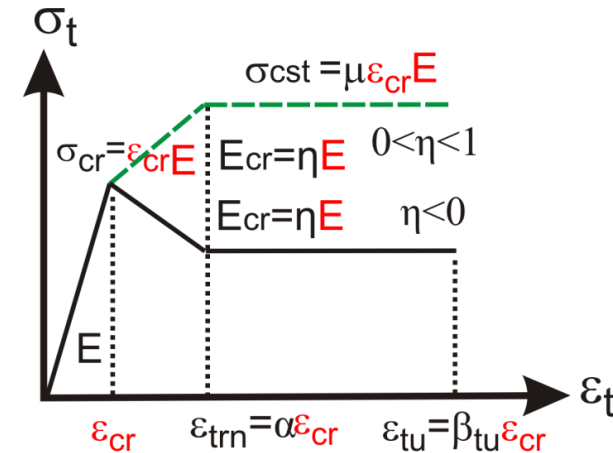


Material Models for Serviceability Design

Compression Model

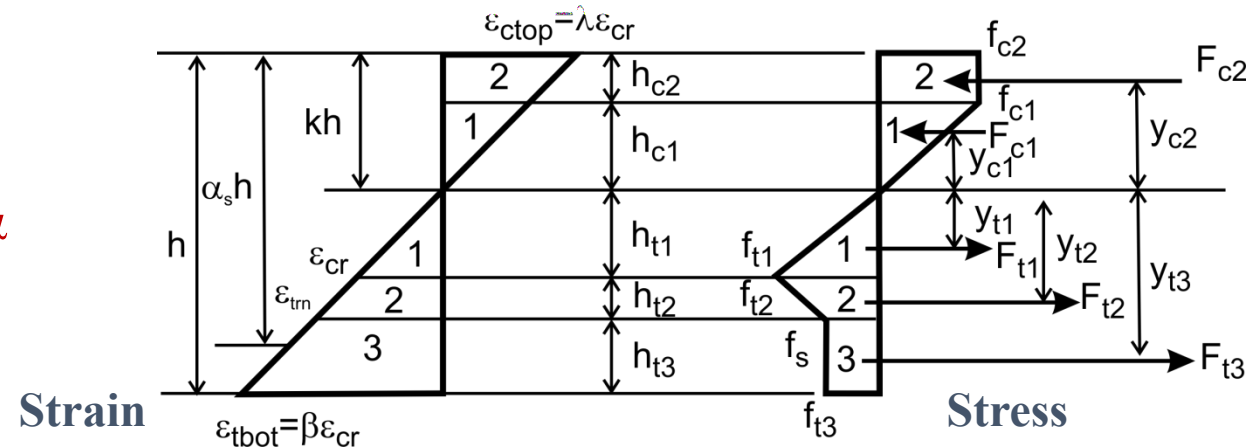


Tension Model

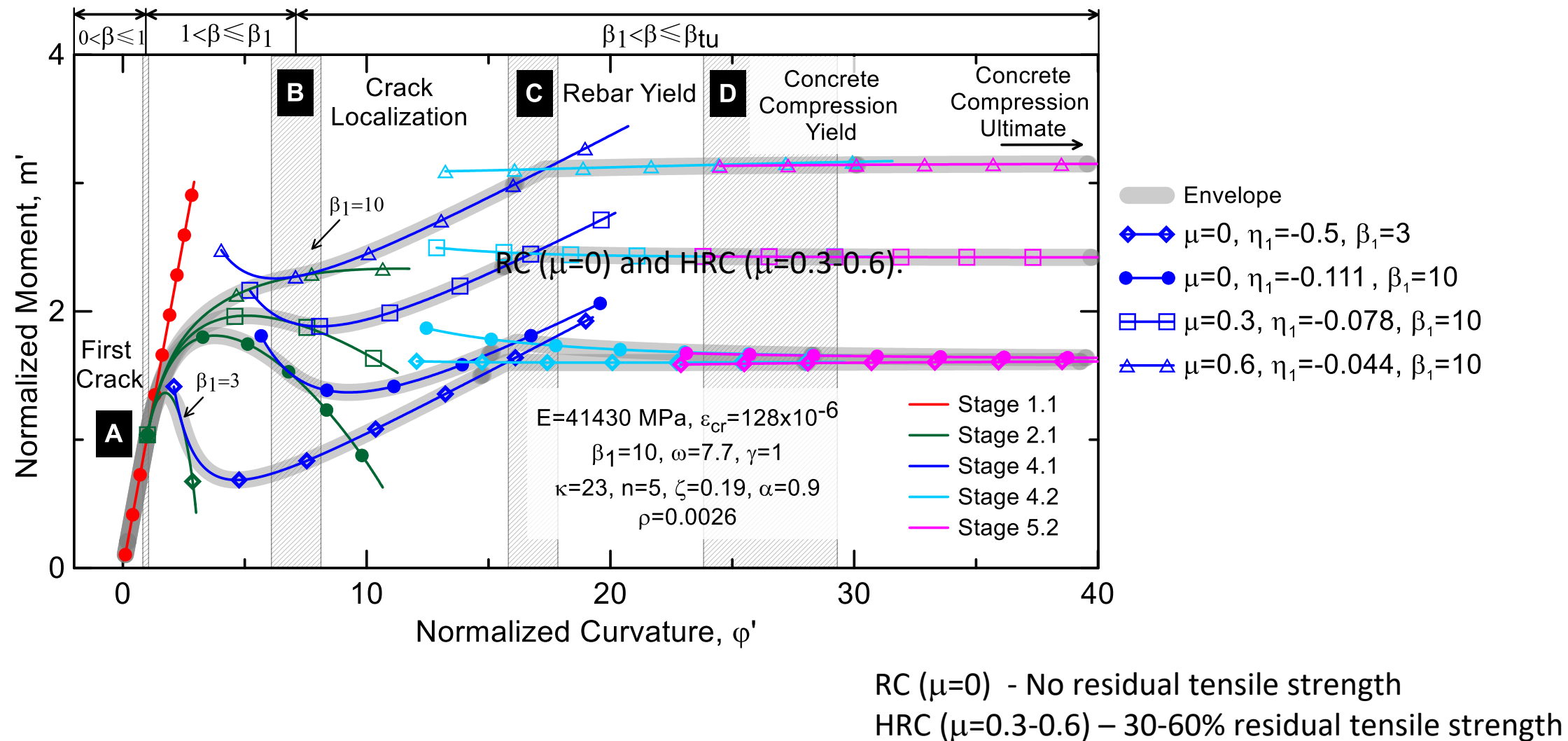


- Serviceability limits: allowable stress, strain, crack width, curvature, deflection

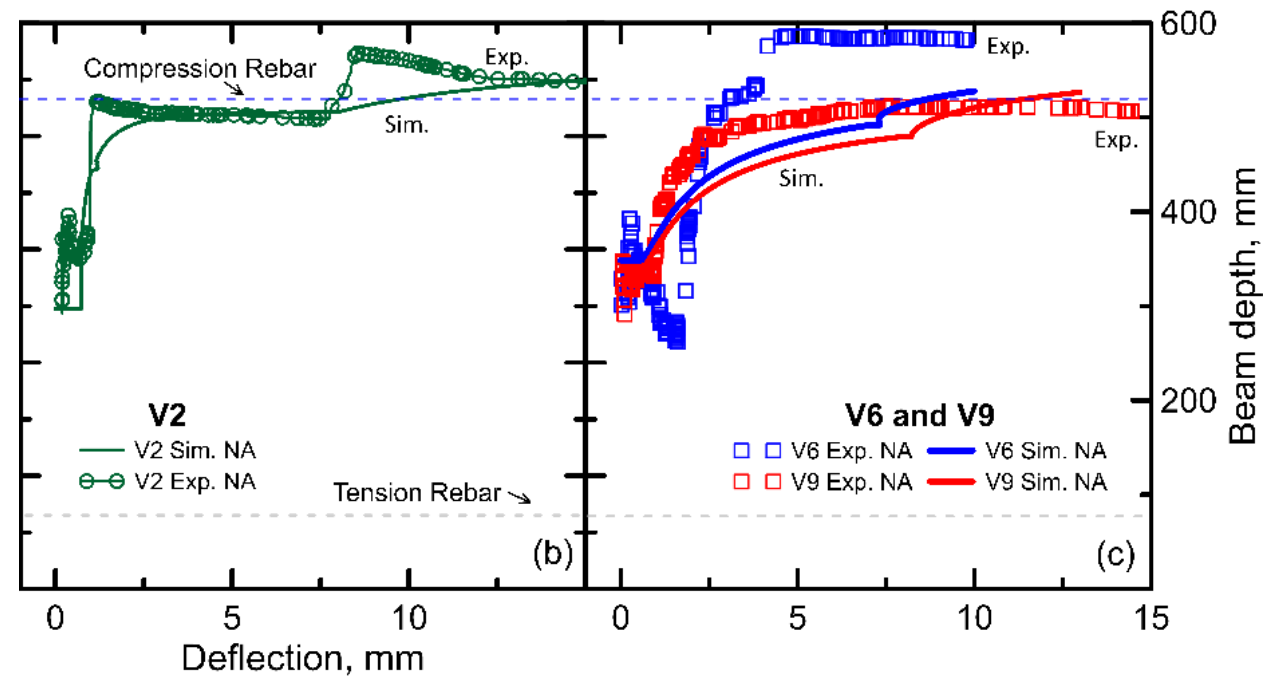
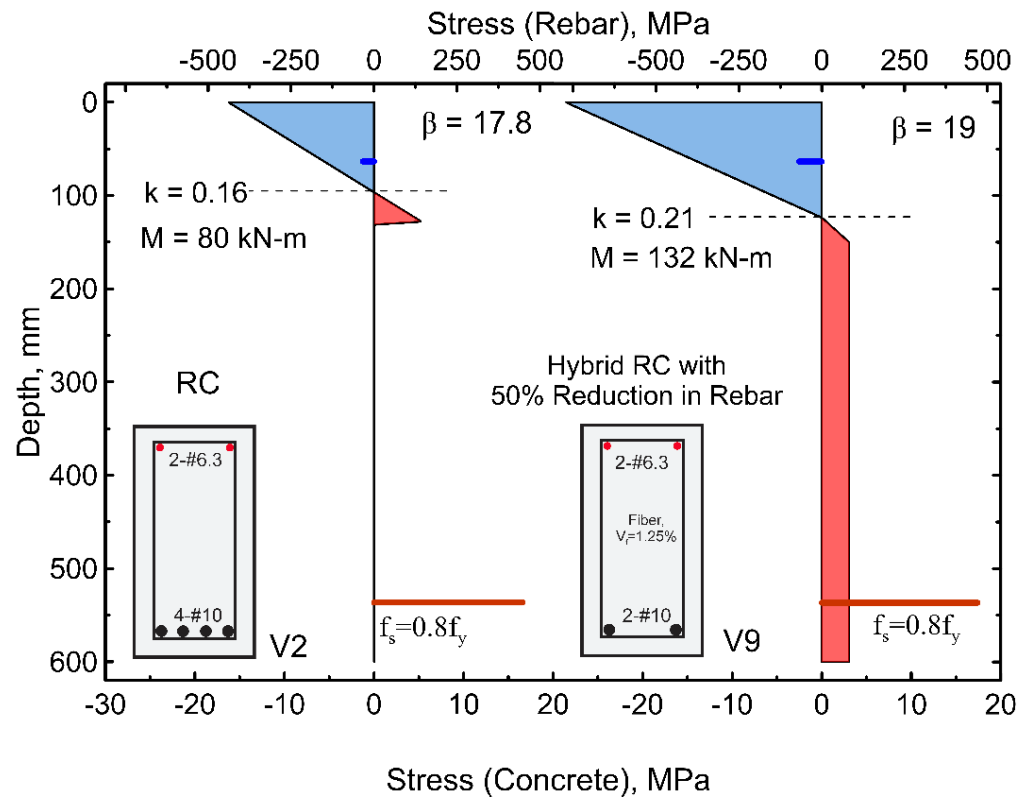
Stage : $\lambda > \omega$, $\beta > \alpha$



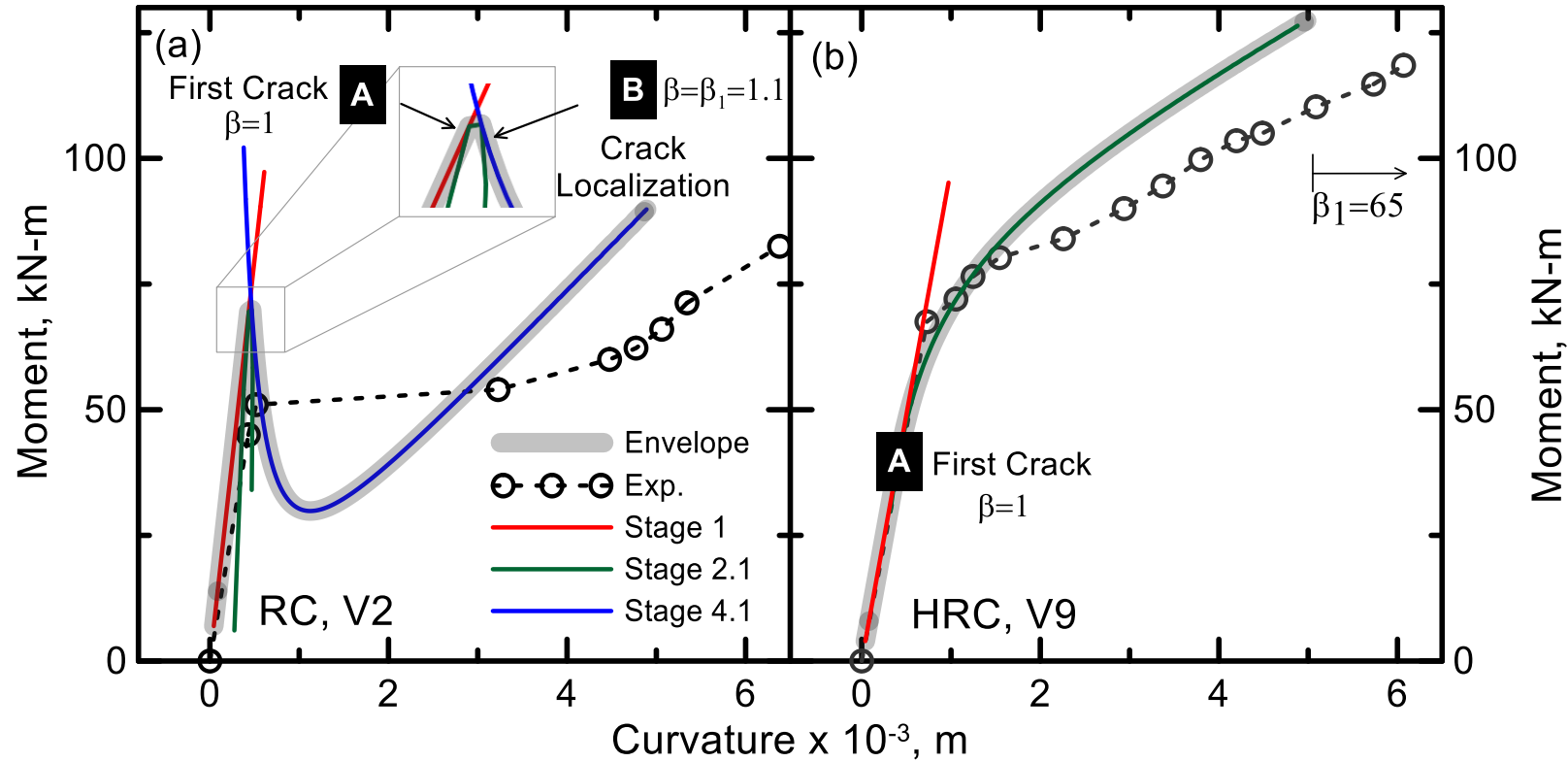
The normalized moment–curvature response with intersection points



Location of Neutral axis and stress distribution

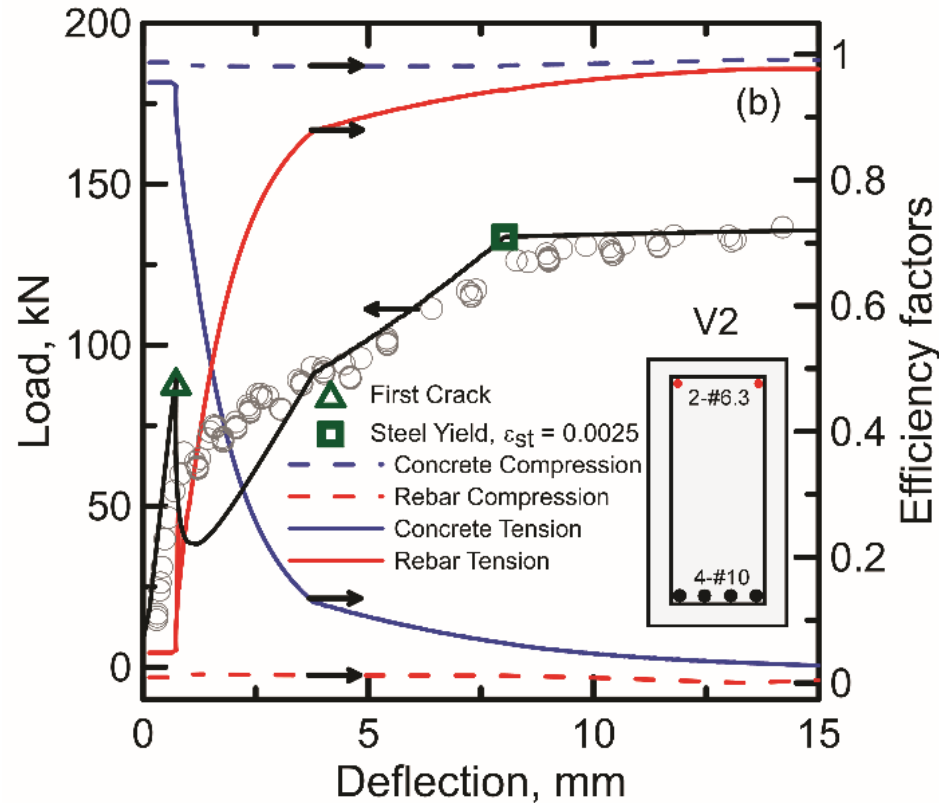


The Localization Phenomenon obtained from the envelope of moment-curvature RC vs Hybrid beam

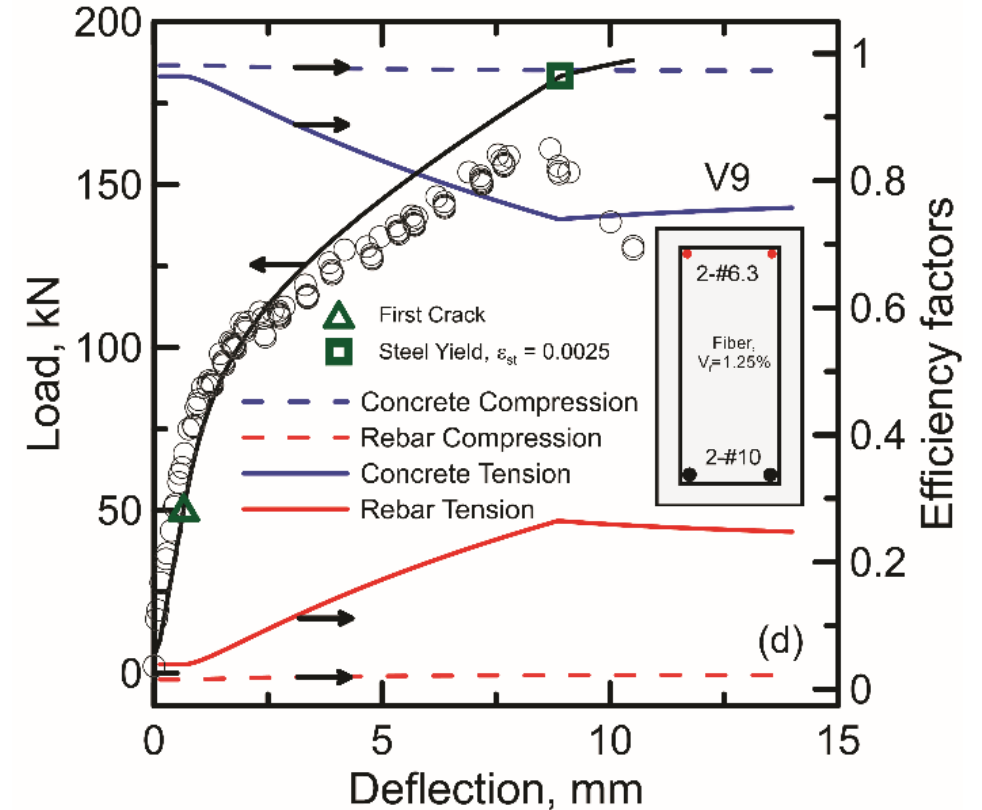


- In conventional RC, Stage 2.1 (green line) is very limited because plain concrete loses tensile capacity immediately after cracking, while the reinforcing bars have not yet elongated enough to contribute tension. This gap causes a significant drop in load until the steel elongates sufficiently to engage.
- The residual tensile strength of the fiber matrix modifies the moment–curvature response by extending the validity of Stage 2.1 before the reinforcing steel becomes active.

Load–Deflection, Failure Interaction, and Efficiency Factors are Linked by a Single Normalized Bottom Strain (β)



$$\eta_{ST}(\beta = \infty) = \frac{F_{ST}}{F_{CT} + F_{ST}} = \frac{n\rho\kappa(\mu + \omega)}{\zeta\kappa n\mu\rho + \kappa n\omega\rho + \mu\omega}$$



$$\eta_{CT}(\beta = \infty) = \frac{F_{CT}}{F_{CT} + F_{ST}} = \frac{(\omega + n\rho\kappa(\zeta - 1))\mu}{\zeta\kappa n\mu\rho + \kappa n\omega\rho + \mu\omega}$$

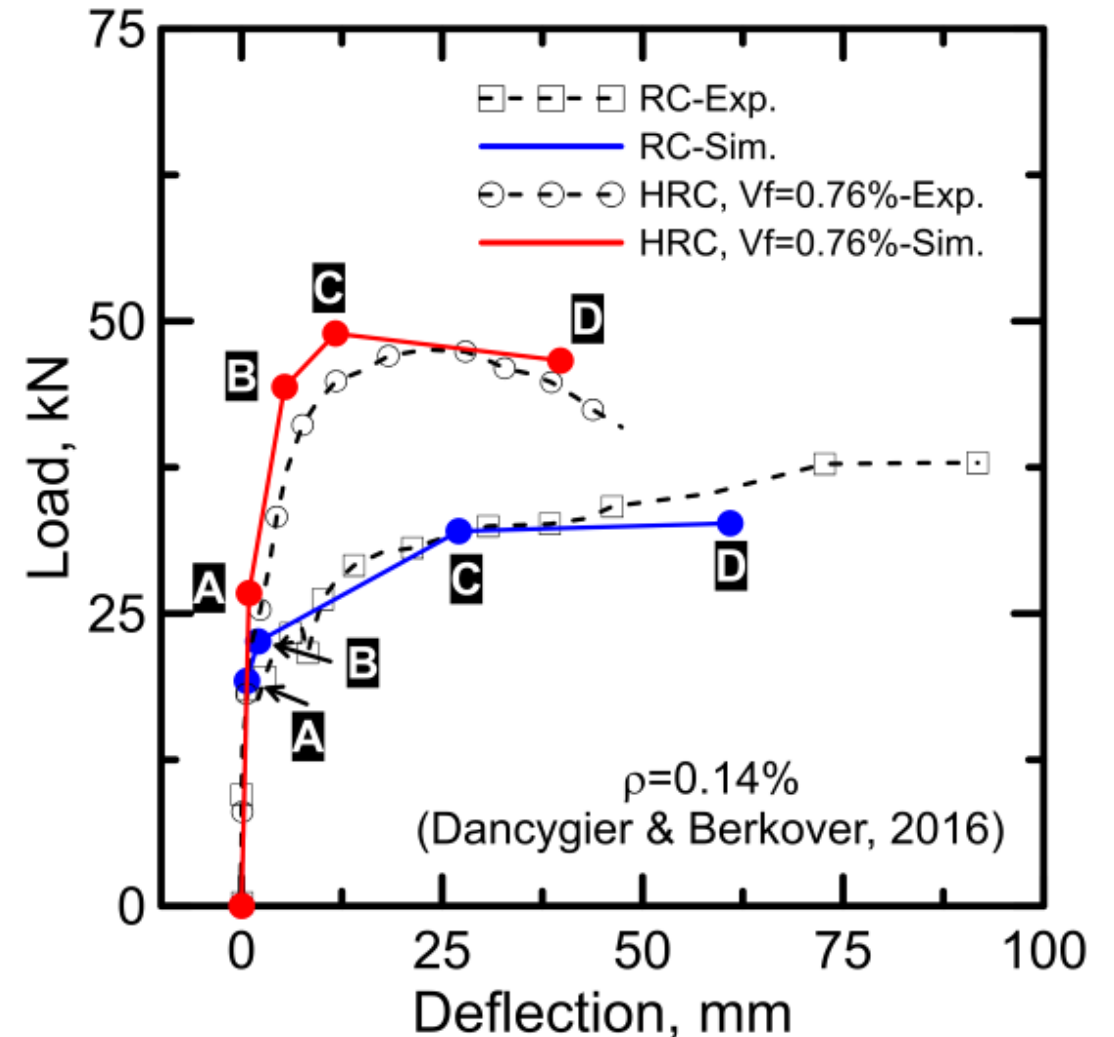
Ductility in Hybrid Reinforced Concrete (HRC)

Conventional RC ductility definition and limitations

- Traditionally, ductility is defined as the ratio of curvature at ultimate failure to curvature at first steel yield (Park, 1988).

Why classical measures can misrepresent HRC?

- In HRC, steel yielding occurs at higher loads and smaller deflections because fibers carry a portion of the tension early on.
- As a result, the classic ductility index may appear lower even when the section shows improved cracking resistance and serviceability performance



Design Procedure for Modelling SHCC Panels for Serviceability

- Material Properties

- Strain hardening material models

- Moment–Curvature Relationship

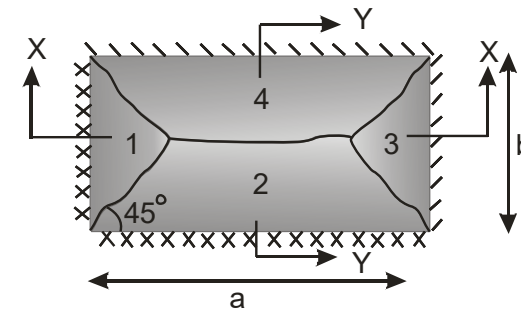
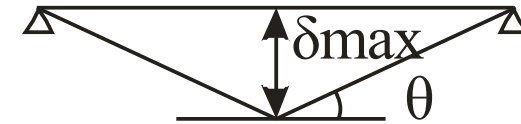
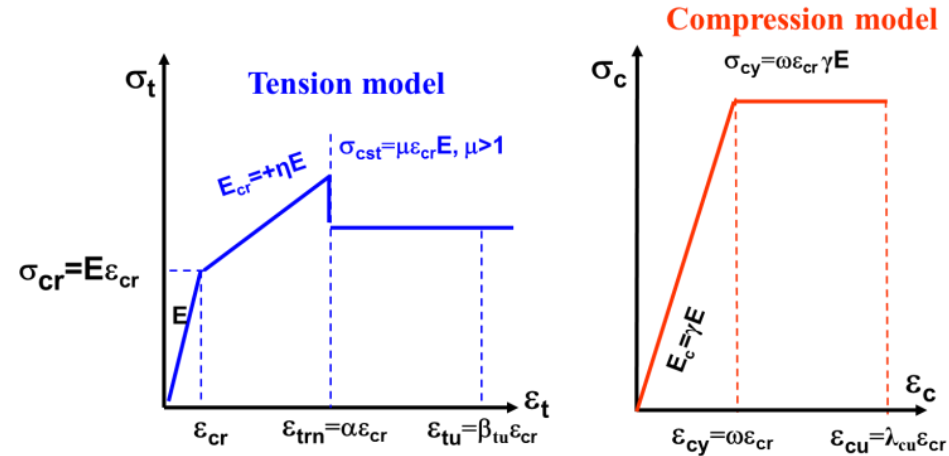
- Cross-sectional analysis

- Curvature-Deflection

- 1-D: Double integration
- 2-D: Kinematically admissible deflections

- Moment-Load , curvature-displacement relationship

- 1-D: Beam statics
- 2-D: Limit State Analysis (Yield Line Theory)



Concluding Remarks

- TRC Composites made with a wide range of textile and fiber types are tailored to meet the performance level specified.
- Parametric material models help in characterization of TRC Systems
- Linearized moment-curvature models used for design of TRC Materials
- Analytical closed form Load-deflection equations of 1-D and 2-D flexural members with FRC, TRC, and HRC
- Using new structural shapes, closed form design equations and a variety of experimental programs, we are able to address the future potential of TRC Structural Materials

Derivation of Moment-Curvature Relationship

Incrementally impose
 $0 \leq \varepsilon_t \leq \varepsilon_{tu}$



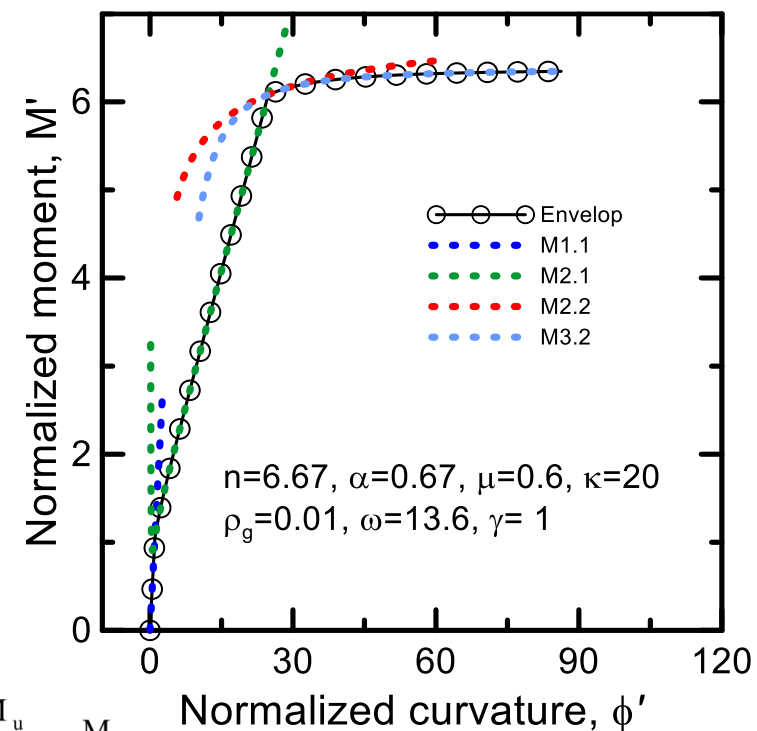
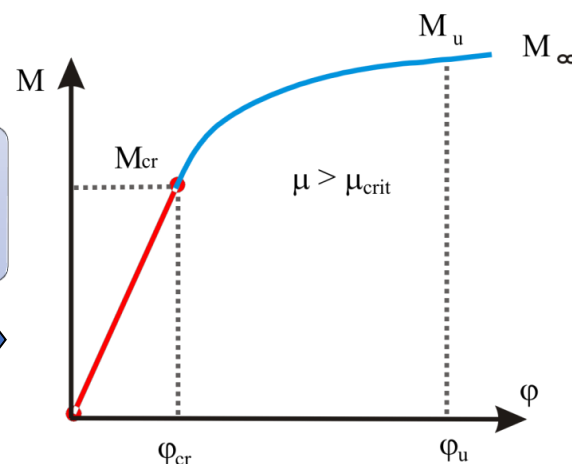
Strain Distribution
 Stress Distribution



$\Sigma F = 0$, determine k



Moment: $M = \Sigma F_{ci}y_{ci} + \Sigma F_{ti}y_{ti}$
 Curvature: $\phi = \varepsilon_c/kh$

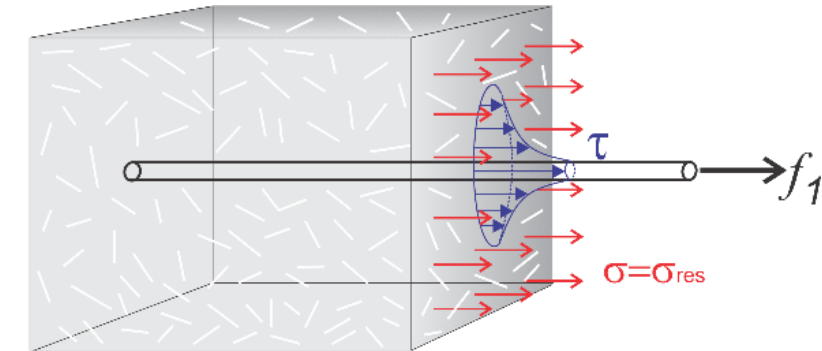
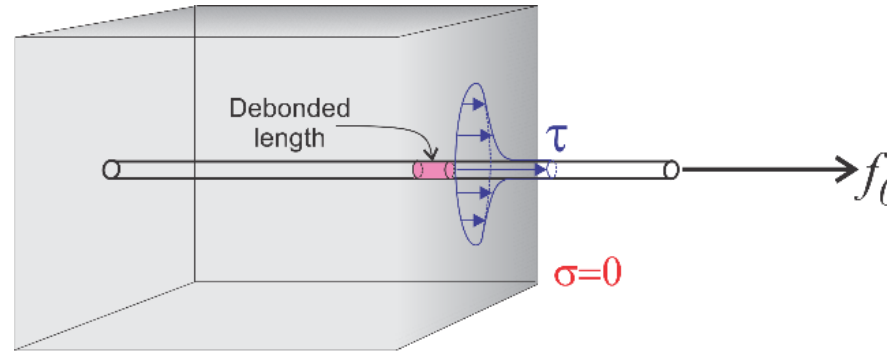


Simplified bilinear
 moment-curvature

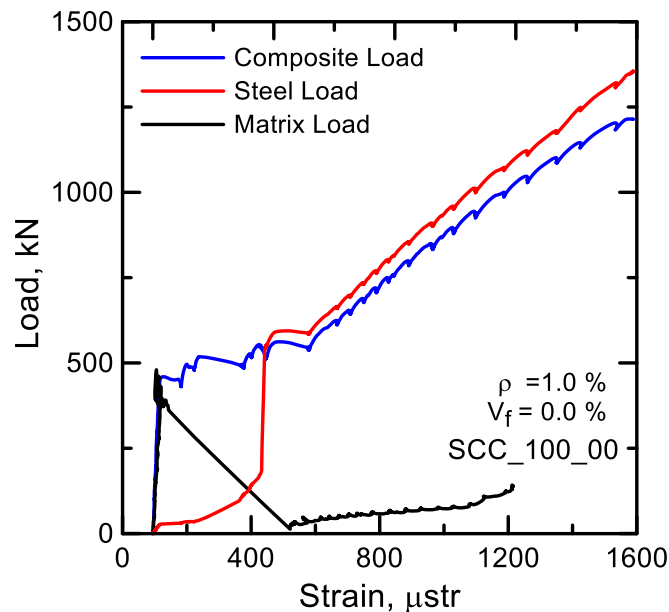


Fiber-Steel Synergy and Post-Cracking Stiffness

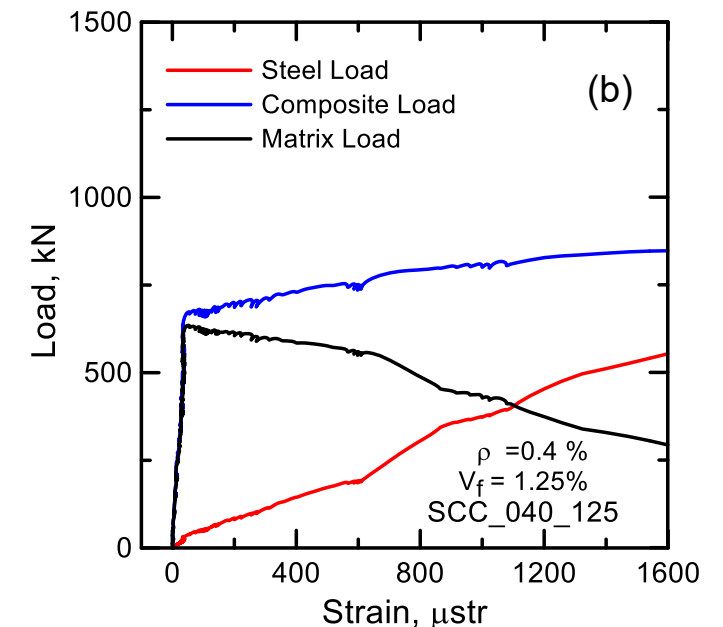
Post-cracking behavior in both tension ties and flexural members is affected by the pullout mechanism. Tension stiffening is a key measure of reinforcement synergy, not captured by the interface



No fiber



With fiber



Fibers reduce bond stress between phases by sharing tensile load and limiting differential strain. This lowers interface shear stresses, slip, crack width and shortens the debonding length,